

# Strong AI Seminar — Review Questions & Answers

Sessions A–L: Singularities, Intelligence, Reality, Time and Space, Information, Action,  
Concept, Memory, Mind, Language, Motion, Brain

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This file collects every question posed in the slide decks (agenda questions, main/secondary questions on individual slides, workshop and repetition prompts, and open “????” questions) and answers each in detail. Organized by session (module): A (Singularities), B (Intelligence), C (Reality), D (Time and Space), E (Information), F (Action), G (Concept), H (Memory), I (Mind), J (Language), K (Motion), and L (Brain). All twelve sessions are complete.

## Session A — Singularities

### Slide 2: Workshop Agenda — “Workshop Singularities”

#### 1. What is a singularity? Define!

A singularity is a point of irreversible, qualitative transition after which the rules, capacities, or trends that previously governed a system no longer hold — borrowed from the mathematical sense of a point where a function becomes discontinuous or undefined. In the technological/societal sense used throughout this seminar, a singularity is not merely an improvement on what came before; it opens an entirely new category of possibility. The deck anchors this with concrete historical examples: mechanization (1769, James Watt’s power machine), electrification (1866, the electrical machine), computerization (1937, Konrad Zuse’s Z1), and the crossover point where more data was stored digitally than in analog form (2002). Each of these didn’t just make existing tasks faster — it created industries, professions, and modes of life that didn’t exist before. Strong AI is proposed as the next candidate in this sequence.

#### 2. Who makes singularities?

Per the deck’s hierarchy (Slide 3), humans — “the Crowning Glory of Creation” — are the originating agents. Humans create machines; machines, work, and work systems (collaboration) are the mechanisms through which singularities are actually produced. So the answer is layered: no single inventor “makes” a singularity alone. It is made by a work system — a network of human creativity, specialized labor (division of labor), tools/machines, and organizational/regional ecosystems (illustrated concretely in Slides 15–16 with the Mainfranken robotics cluster) that together turn an invention into a world-changing transition. As machines and AI systems become more capable, they may increasingly participate in generating future singularities themselves, rather than being purely passive instruments.

#### 3. How are singularities made?

Singularities are made cumulatively, not spontaneously. Slide 6’s technology timeline shows each wave of change providing the infrastructure for the next: mechanization (power machines) enables electrification (electrical machines, dynamos), which enables computerization (Zuse’s Z1, transistors), which enables semiconductors, integrated circuits, networks, sensors, IoT, and Big Data — which in turn create the substrate on which Strong AI becomes possible. Practically, the mechanism has four ingredients: (1) a core enabling invention, (2) infrastructure/scaling so the invention can be mass-produced and distributed, (3) social and legal adaptation (“Acceptance, Legal Issues”) so the technology can be integrated into daily life, and (4) organized work systems (division of labor, regional industrial ecosystems) that translate the invention into widespread economic and social impact.

#### 4. Why are singularities made?

Because humans are defined (Slide 4) as agents whose basic drive includes the desire to “Simplify Work” — singularities are, at root, attempts to remove friction, effort, or limitation from the tasks that have to be done. This is reinforced by Slide 7, where “Division of Labor” and the connective role of work explain why change compounds: once one bottleneck is resolved (e.g., mechanical power), the next constraint (e.g., information processing, or now general intelligence) becomes the new target for innovation. Secondary motives include competitive advantage (regions/companies racing to be early adopters, Slide 15–16), curiosity, and the general human tendency toward creativity as a marker of intelligence.

#### 5. Give examples of huge/primary singularities

- Mechanization — 1769, James Watt’s power machine (steam engine)
- Electrification — 1866, the electrical machine (plus Werner von Siemens, dynamos)
- Computerization — 1937, Konrad Zuse’s Z1; 1948, the transistor (Shockley, Bardeen, Brattain)
- The digital/analog data crossover — 2002 (more data stored digitally than in analog form)
- Strong AI / AGI (“Thinking Machine,” “Humanoid Machine”) — presented as the anticipated next primary singularity

## 6. Give examples of small/secondary singularities

- Integrated circuits (ICs) and semiconductor miniaturization
- Networks, sensors, IoT, and Big Data as enabling infrastructure layers
- The automotive → “robomotive” industry transformation (Slides 15–16): a sector-specific disruption rather than a civilization-wide one
- Regional industrial ecosystems (e.g., the Mainfranken robotics supply chain of gears, bearings, controllers, electric machines) that adapt early and gain localized advantage

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### Slide 4: Humans and Human Abilities

**Main question: What are Humans?** Answer given and expanded: **Working Individuals** — the deck deliberately defines humans functionally rather than metaphysically. Rather than starting from consciousness or identity, it starts from agency and productivity: a human is a being that continuously does things — physical and mental work — and that fact is treated as the most useful starting point for comparing humans to machines later in the argument.

**What are Humans Basically Able to do? Move & Think.** These are presented as the two irreducible human capacities: physical/motor agency (embodiment, action in the world) and cognitive processing (reasoning, perception, planning). Any serious model of “strong AI” must therefore account for both dimensions — a system that only thinks (a pure reasoning engine) or only moves (a pure actuator) is incomplete relative to the human baseline.

**What is a Human’s Basic Instinct?** The slide leaves this implicit, but in context it is best answered as: the drive toward purposeful action and self-preservation, expressed through continuous work and creative problem-solving. This connects directly to the next bullet — humans don’t just move and think passively; they are driven to act in order to close the gap between a current state and a desired one (a theme that resurfaces explicitly in later modules on Action and Information, where “knowledge gaps” trigger motivation).

**Are Humans Intelligent? Yes** — but the answer is deliberately practical, not philosophical. Intelligence is evidenced by two concrete markers: **Creativity** (the ability to generate novel solutions or ideas) and **Simplify Work** (the ability to reduce friction, effort, or complexity in performing tasks). This is significant because it defines intelligence operationally/functionally (what it does) rather than through abstract properties like self-awareness — a definition that is far easier to compare directly against machine capabilities.

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### Slide 5: Machines and Their Role

**Main question: What is a Machine?** A machine is defined functionally as a **Work Performer** — any constructed system whose purpose is to execute work. This mirrors the human

definition (“Working Individuals”) deliberately, setting up a direct human-vs-machine comparison on the same functional axis: both are ultimately judged by what work they perform, not by their internal nature.

**What Kinds of Machines Exist?** Two categories: **Mechanic** (machines that perform work via physical movement and applied force — engines, motors, mechanical devices) and **Non-Mechanic** (machines that perform work without traditional moving parts — software, algorithms, digital/AI systems). This distinction matters because it shows that “machine” is not limited to hardware with gears and levers; a purely computational/software system counts as a machine under this framework.

**What Kind of Work can a Machine Perform? Physical & Mental.** This is the pivotal claim of the module: machines are not restricted to brute-force, repetitive physical labor. They can also perform cognitive/computational tasks — calculation, logic, pattern recognition — which is precisely the opening that makes a “strong AI” argument possible. If machines can only ever do physical work, there is no basis for comparing them to human mental capability; by asserting machines can do both, the deck sets up the rest of the seminar’s central question (can machines eventually match or exceed human mental work at a general level?).

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## Slide 7: Work as Connection

**Main question: What is Work? “Things that have to be done.”** Deliberately broad — work is any necessary or purposeful task, encompassing both obligatory activity and the general processes that drive systems and change forward.

**What Types of Work Exist? Physical & Mental** — the same split used for humans and machines, reinforcing that both bodies and brains (biological or artificial) contribute to productive activity, and that any comparison between human and machine capability must consider both dimensions.

**Chances of Work?** Interpreted as: work is not purely a burden — it represents opportunity. Performing tasks is also where innovation, improvement, and choice arise; the “chances” of work are the possibilities for value creation, efficiency gains, and new capability that emerge from doing things rather than merely enduring them.

**What Makes Work so Special? Division of Labor.** Specialization and cooperation among distinct workers (human, machine, or both) is the mechanism that allows complexity and progress to emerge — no single agent needs to do everything, so systems can scale far beyond individual capability. The slide’s concluding statement — that work “**Connects People and Machines**” — extends this: work is the shared interface or medium through which human intent and machine execution meet, which is why singularities are best understood not just as new machines appearing, but as a restructuring of how work itself is organized, divided, and connected between human and machine actors.

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## Slide 3 / Slide 13: Open Questions on Strong AI (“?????”)

**Is Strong AI the “Next Singularity”?** *(Slide 3, marked with a question mark)*

Posed as an open hypothesis rather than a settled claim. Based on the deck’s own later framework (Slide 14), Strong AI would need to simultaneously master Intelligence, Memory, Mind,

Reality, Time and Space, Information and Knowledge, Language, Motion, Action, and Conceptualization — not just one isolated capability like data storage (computerization) or communication (electrification/networks) the way prior singularities did. That breadth is why it is presented as the strongest candidate for the *next* primary singularity: it would not just extend one axis of capability but potentially replicate the whole integrated cognitive-and-embodied capacity set that, until now, only humans possessed. The question mark signals the presenter’s honest uncertainty about whether this is inevitable or merely one possible future among others.

**Is AGI “Same as Strong AI”?** (*Slide 13, marked “????”*)

Treated as broadly synonymous in casual usage, but the question marks flag a real unresolved ambiguity in the field. **Artificial General Intelligence (AGI)** typically emphasizes *generality* — the ability to perform well across a wide range of domains and tasks, without being restricted to one narrow specialty. **Strong AI**, in its original (Searle-derived) philosophical sense, emphasizes *genuine understanding or consciousness* rather than mere simulation of intelligent behavior. A system could in principle be general (AGI) without possessing genuine understanding (not “strong” in the philosophical sense), or a narrowly-scoped system could arguably exhibit some form of “strong,” conscious-like processing without being general. The deck deliberately leaves this open so the audience interrogates the equivalence rather than accepting it uncritically.

**Can machines achieve “Universal Learning and Acting like Humans”?** (*Slide 13, marked “????”*)

This is presented as the defining aspiration of Strong AI, explicitly not yet an achieved fact — hence the question marks. Per the slide’s own taxonomy, reaching it would require unifying three currently separate AI traditions into one domain-general learner: **Physical AI** (embodied, spatial action in the real world), **Discriminative AI** (classification and prediction), and **Generative AI** (creativity, producing novel content). Current systems — including today’s large generative models — are strong within one or two of these categories but do not yet combine all three into a single system that learns and acts across arbitrary domains the way a human does. The honest answer, consistent with the deck’s framing, is: not yet, and this is the open engineering and philosophical challenge the rest of the seminar (Modules B–L) is built to address.

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## Session B — Intelligence

### Slide 2: Workshop Agenda — “Workshop Intelligence”

#### 1. What is intelligence? Try to give a definition.

Per the Duden dictionary definition the deck cites directly (Slides 4–5): intelligence is “*the ability [of human beings] to think abstractly and reasonably, and to derive purposeful action from that.*” In the engineering terms the module builds toward, this collapses into a three-stage pipeline: **abstract/rational thinking** → **reasoning** → **purposeful, goal-directed action**. Crucially, the definition judges intelligence by output — what an agent *derives and executes* — not merely by what it stores or knows. This is also the yardstick the rest of the module measures machines against.

#### 2. What is needed to be intelligent?

Per Slide 3’s three foundational axioms, intelligence requires all three of the following simultaneously — none is optional: - **Interaction** — “No Interaction, no Intelligence!” An agent must sit inside a closed perception–processing–action loop with an environment (Slide 7: Impressions → Perception/Recognition → Solving/Learning → Motion/Speech). - **Learning** — “No Learning, no Intelligence!” The agent must be able to gain insight and knowledge from that interaction, not just store data. - **Thinking** — “No Thinking, no Intelligence!” The agent must be able to process impressions, memories, and experience into insights and plans. Slide 12 turns this into a scorecard: Information, Action, and Thinking must all be present and mature — the deck rates Information as fully solved (+/+) but Action and Thinking as only partially solved in machines (+/+/-), meaning “being intelligent” today is only partially achieved by machines.

#### 3. Give examples of “weak” intelligence.

Narrow, single-domain AI systems: image classifiers, recommendation engines, domain-limited chatbots, predictive/statistical models — anything built through conventional **Machine Learning**. Slide 8 is explicit that Machine Learning is *not* general learning: it is curve-fitting and pattern optimization within a fixed objective and domain, useful and real, but structurally incapable of the cross-domain, self-directed cognition the deck reserves for “strong” intelligence.

#### 4. Give examples of “strong” intelligence.

Human general intelligence is the working example throughout: the ability to reason abstractly across arbitrary new domains, learn adaptively without being purpose-built for a task, form plans, and act with intrinsic motivation. In machine terms, this is the not-yet-realized goal of **Artificial General Intelligence (AGI)** — explicitly confirmed in Slide 14 to *not yet exist* (“Does AGI already exist? No.”) but judged theoretically achievable (“Is AGI Possible? Yes.”). It would require the full, unified cognitive map of Slide 10 (language, concept, information, the three memory types, reality, mind, space, time, orientation) operating together, not any single capability in isolation.

#### 5. Do you need to know the concept of intelligence to be intelligent?

No — the deck treats this as a genuine philosophical hook rather than a rhetorical question. It is entirely possible to *exhibit* intelligent behavior (solving problems, adapting, learning) without possessing any conceptual, self-reflective grasp of what “intelligence” is — animals behave intelligently with no theory of intelligence, and today’s narrow AI systems solve real problems with zero self-model. This matters because it decouples the *exercise* of intelligence (behavior, output) from *meta-cognition about* intelligence (the concept) — a system can satisfy the Duden/operational definition of intelligence without any philosophical self-awareness at all.

## 6. What is thinking? Try to give a definition.

Per Slides 9 and 11: thinking is “**the base of intelligence**” — the process of processing impressions, memories, and experiences in order to produce insights and plans. The deck’s formal, engineering-grade definition (Slide 11) is: “*the processing of all impressions for using them for the planning and execution of goal-oriented acting (expressions), after the need for action was determined.*” Broken into stages: (1) intake all sensory impressions, (2) evaluate whether a need for action actually exists, (3) plan using stored knowledge/memory/experience, (4) execute that plan as goal-oriented action or expression.

## 7. How is intelligence and thinking connected together?

Thinking is the *mechanism*; intelligence is the resulting *capability/state*. The deck builds an explicit dependency chain: Thinking is what makes Learning possible (Slide 8: “How does Learning Work? → By Thinking”), and Learning is one of the three non-negotiable requirements for Intelligence (Slide 3’s axioms). So the chain runs **Thinking → Learning → Intelligence**, with Thinking also directly generating the “purposeful acting” the Duden definition requires. In short, intelligence cannot exist without thinking because thinking is the active, generative process — intelligence is essentially the demonstrated capability that successful thinking produces.

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### Slide 3 / Slide 6: “What is Artificial?”

Per Slide 6, “Artificial” is defined through a structural dichotomy rather than a single sentence: **Natural Intelligence** (rooted in the biological world — body, spirit/mind, rationality operating on a physical, biological substrate) versus **Artificial Intelligence** (rooted in the digital world — the same properties of cognition, rationality, and mind, reconstructed synthetically in silicon/software). The key implication: “artificial” should not be read as “fake” but as a **digital reconstruction of biological cognitive processes**. For genuine parity with natural intelligence, the digital world has to find synthetic substitutes for embodiment (e.g., physical robotics or simulated environments) — a disembodied digital mind without a “body” analogue is, per this framework, an incomplete reconstruction.

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### Slide 3 / Slides 13–14: “Already strong or general or not? What to do?”

**Is current AI already strong or general?** No. Per Slide 13’s maturity timeline, present-day systems sit firmly in the **Weak AI / “Presence”** category: specialized, narrow, and lacking unified cognitive synthesis. Slide 14 confirms this directly: “Does AGI already exist? → No.” Strong AI/AGI is placed on the “**Future**” side of the timeline — an unachieved engineering target, not a current reality (despite market hype).

**What to do about it?** Per Slide 12’s capability scorecard, the gap is not in **Information** (fully solved theoretically and practically, +/+ ) but in **Action** and **Thinking** (both rated +/- — partial, immature in machines). The path forward, per Slides 12–13, is to progressively master and *unify* the interaction-loop components (Perception, Recognition, Solving, Learning, Speech/Motion) rather than wait for one singular breakthrough algorithm — moving each faculty from narrow, task-specific implementations toward integrated, general-purpose execution.

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## Slide 4: Defining Intelligence

**Can we Build Machines as Intelligent as we are?** (*posed with a question mark, left open*)

The deck deliberately leaves this open in Slide 4, then partially resolves it by the end of the module (Slide 14): machine general intelligence does **not currently exist**, but is judged **theoretically possible**. So the honest, deck-consistent answer is: *not yet, but in principle, yes* — the obstacle is engineering the full “thinking” mechanism (Slides 9–12), not any fundamental, in-principle impossibility.

**Can we Promote Intelligence?** → **Yes.**

Yes — intelligence, at least in its operational/algorithmic sense, is not a fixed, static trait; it can be trained, optimized, and cultivated. This is the pragmatic justification for machine learning and systematic engineering approaches: even without solving full AGI, the intelligence exhibited by existing systems can continuously be improved and “promoted” incrementally.

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## Slide 8: Learning

**What is Learning?** → **Gaining Insights and Knowledge.**

Learning is defined as the *active acquisition of insight and knowledge* — explicitly not passive data accumulation or storage.

**How does Learning Work?** → **By Thinking.**

The mechanism behind learning is thinking, an active, evaluative process — not merely statistical optimization. This upends the standard machine-learning picture, where models “learn” via gradient descent and backpropagation over data with no evaluative “thinking” step; the deck implies that generic, adaptive, human-like learning instead requires an active, top-down cognitive process to interpret and judge what is being experienced.

**Is Machine Learning General Learning?** → **No.**

Contemporary Machine Learning is explicitly denied the status of “general learning.” It is narrow pattern-fitting and mathematical optimization within a fixed objective and domain — not the holistic, adaptive, thinking-driven learning natural intelligence exhibits. Despite this limitation, Machine Learning is still affirmed as “**the basis for AI**” — the necessary technical foundation on which higher cognitive architectures (true thinking machines) must eventually be built; it’s a required starting point, just not a sufficient end point.

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## Slide 9: Thinking

**What is Thinking?** → **The Base of Intelligence. Processing Impressions, Memories and Experiences.**

Thinking is defined as the foundational, dynamic process underlying intelligence: it takes three distinct inputs — immediate external impressions, stored internal memories, and accumulated experience — and synthesizes them together.

**What is Thinking doing?** → **Producing Insights and Plans.**

The literal function/output of thinking is to generate insights (new understanding) and plans (structured courses of future action) — directly connecting back to the Duden definition’s requirement of “deriving purposeful acting.”

**How does Thinking Work?** (*left textually unanswered on Slide 9 — a deliberate cliffhanger*)

The mechanics are answered structurally across the following slides: - **Slide 10** maps thinking as a network of interconnected cognitive faculties rather than one process — Language, Concept, Information, Knowledge, Episodic/Perspective/Value Memory, Reality, Mind, Mental Eye, Space, Time, and Orientation, all cross-talking rather than functioning as isolated databases. - **Slide 11** gives the formal four-stage pipeline: (1) intake all impressions, (2) evaluate whether a need for action exists, (3) use impressions/memory/experience to plan, (4) execute that plan as goal-oriented action/expression.

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**Slide 12: “What we need to build a thought machine”**

| Core Cognitive Component | Theory | Machine |
|--------------------------|--------|---------|
| Information              | +      | +       |
| Action                   | +      | +/-     |
| Thinking                 | +      | +/-     |

**Answer:** Building a genuine thought machine requires three components in combination: **Information** (data storage/transmission/retrieval — theoretically mature *and* practically solved in machines today), **Action** (execution in physical or digital space — theoretically understood but only partially achieved in machines: robotics and autonomous digital action across unstructured environments remain unsolved engineering problems), and **Thinking** (the cognitive synthesis mapped in Slides 10–11 — theoretically well-framed by cognitive science and philosophy, but machines today only achieve a fractional version: narrow pattern prediction rather than generalized synthesis and intrinsic planning). The bottlenecks blocking AGI are therefore not data-related but lie specifically in mastering **Action** and **Thinking** and integrating them symmetrically with Information.

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**Slide 14: Conclusion — The Core Realities of AGI**

**Is Machine Thinking complex? → Yes. Much more than Machine Learning.**

Yes. Machine thinking — the full, multi-nodal cognitive network mapped in Slide 10 (spanning language, concept, information, three distinct memory types, reality, mind, space, time, and orientation, all interacting) — is orders of magnitude more complex than Machine Learning’s narrow statistical pattern optimization. ML is one supporting basis/component of the picture; thinking requires dozens of interacting cognitive faculties operating as a unified system.

**Does AGI already exist? → No.**

No. The deck flatly rejects current marketing narratives and public misconceptions: today’s systems remain “Weak AI” — narrow and specialized — and per Slide 12’s scorecard, only partially (+/-) fulfill the Action and Thinking requirements needed for true general intelligence.

### **Is AGI Possible? $\rightarrow$ Yes.**

Yes. Despite its current absence and immense complexity, the presenter maintains an optimistic, engineering-grounded stance: building a machine capable of true general intelligence is scientifically and structurally possible — not a mystical impossibility, but a profound, long-term engineering goal contingent on fully building out the Interaction, Learning, and Thinking axioms established in Slide 3, together.

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## Session C — Reality

### Slide 2: Repetition — “Singularity” (Review of Module A)

**Which technological unique developments (singularities) massively changed our daily life and work?**

Mechanization (1769, James Watt’s power machine), Electrification (1866, the electrical machine), Computerization (1937, Konrad Zuse’s Z1; 1948, the transistor), the digital/analog data crossover (2002), and the anticipated Strong AI/AGI singularity — see *Session A* for full detail.

**Which probably will lead to the humanization of machines?**

Per Module A’s technology timeline, the social/cognitive track — running through **Electronization**, **Intellectualization**, and **Emancipation** toward **Humanization** — rather than the raw hardware track (mechanization, electrification). This is the trajectory that gives machines increasingly human-like traits and cognitive autonomy, culminating in the “Humanoid Machine” and “Thinking Machine” concepts: Strong AI/AGI development and humanoid robotics are what is expected to actually humanize machines, not further gains in raw computing power alone.

**Why is “Work” as a concept so important?**

Because Work is the connecting medium between humans and machines (Module A, Slide 7 & 11): it is the interface where human intent meets machine execution, structured by the division of labor, and it is the underlying activity from which singularities themselves emerge — singularities are best understood as restructurings of how work is organized and shared between human and machine actors.

**What components make up a “Work System” and how are the components connected?**

Per Module A, Slide 12: **Work Task** (why), **Worker(s)** (who), **Work Tool(s)** (with), **Work Process** (how), and **Work Object(s)** (what). They connect through **Work** itself, placed at the center as the mediating concept — purpose (why) is carried out by an actor (who), using a tool (with), following a process (how), to produce or transform an object (what).

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### Slide 3: Repetition — “Intelligence” (Review of Module B)

**Give a proper definition of intelligence.**

Per the Duden definition used throughout Module B: *“the ability [of human beings] to think abstractly and reasonably and to derive purposeful action from that.”* See *Session B* for the full breakdown of this three-stage pipeline (abstract thinking → reasoning → purposeful action).

**Which components does a machine need, to be intelligent and human like?**

Per Module B’s axioms (Slide 3) and scorecard (Slide 12): **Interaction** (a closed perception–processing–action loop with an environment), **Learning** (gaining insight/knowledge, driven by thinking), and **Thinking** (processing impressions, memories, and experience into insights and plans) — formalized as the three pillars **Information + Action + Thinking**, all of which must mature together, not in isolation.

**What is the only thing humans can do?**

This is posed as a provocative, unanswered pivot in the slide itself, but the module’s own interpretive framing (and Module A’s “Crowning Glory of Creation” motif) points to a specific answer: humans are, so far, uniquely able to **originate genuinely novel purpose and meaning from raw reality** — to set the initial goal, the “why,” from nothing, rather than execute, optimize, or recombine within a framework someone or something else already defined. Even generative AI produces novelty by recombining patterns learned from prior human-generated data; it does not (yet) independently decide that a wholly new class of problem is worth solving. This “prime mover” role — initiating purposeful work systems and generating original conceptual models out of undifferentiated reality — is the boundary the rest of the module (and the eventual “Reality” framework) is built to help machines eventually cross.

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## Slide 4: Workshop “Reality” — the five core agenda questions

### 1. What is reality? Define!

Per Slide 8’s formal answer, reality is “**what surrounds us with ourselves included**” — the totality of the physical (matter and waves) and mental (minds, thoughts, representations) domains, all contained within a single, shared **World**. Crucially, reality is not just “external stuff”: the observer is embedded inside it, and minds/thoughts are themselves part of it, not a separate mystical layer.

### 2. How do Computer Scientists handle reality?

Traditionally, through object-oriented modeling, relational database schemas, and structured data representations that treat objects as isolated, self-contained instances. The module argues this is fundamentally too narrow — real entities cannot be modeled accurately without accounting for their embeddedness in a larger “world” container (Slide 13), their multiple ontological categories (physical/mental, matter/wave, per the Slide 9 matrix), and the fact that every observer holds its own distinct, non-identical **Representation** (Slide 24) rather than a single universal data copy.

### 3. Why does a digital brain not yet exist, even after half a century of effort by hundreds of thousands of computer scientists?

Because the bottleneck isn’t time, manpower, or lack of effort — **Reality itself is extraordinarily difficult to map objectively into computational logic**. Per the rest of the module, prior software modeling failed to capture: the layered, observer-dependent nature of representation (Primary Reality vs. Secondary/n-ary Reality, Slides 27–35); the “Sandbox Principle” that a mind never perceives an entity directly, only its own internal simulation of it (Slide 25); and the underlying **Paradox and Dualism** of mind and matter (Slide 38). A true digital brain needs an architecture that unifies World, Entity, Differentiation, Representation, Primary Reality, and n-ary Reality simultaneously — a systemic integration that 50+ years of narrower, siloed engineering approaches have not achieved.

### 4. Is reality subjective or objective?

Both, depending on the layer, per the module’s own resolution: **Primary Reality** (the shared physical world) does have Objective Truth (Slide 29) — a single, verifiable ground state — even though every mind’s **Representation** of it is inescapably subjective and unique (no two minds share an identical model, Slide 29). At the **n-ary** level (Slide 32), by contrast, there is explicitly **no** objective truth — only shared agreements (money, law, language) that hold only

as long as collective belief in them persists. So: the physical substrate is objective, individual representations of it are subjective, and symbolic/social systems built on top are intersubjective (agreement-based).

## 5. Is there one or are there more realities?

Both. There is exactly **one** shared **Primary/physical World** (Slide 8: “Is there one world? Yes. We all face the same one world.”) — but there are as many individual **Representations** of it as there are minds (over 8,000,000,000, per Slide 24), plus a recursively nested hierarchy of **Secondary, Tertiary, Quaternary, and n-ary realities** (Slides 32–35) built on top of that single physical base, as agents model each other’s models of the world.

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### Slide 8: World Fundamentals

**What is the World? → What surrounds us with ourselves included.** Defines the world as an all-encompassing container rather than an external stage the observer stands outside of — the observer is embedded within it, not a detached spectator.

**Are Minds and Thoughts Part of the World? → Yes.** Mental activity is not a separate mystical dimension; it is an objective phenomenon occurring inside the one physical universe. This claim is what the entire module’s closing “Paradox and Dualism” (Slide 38) is built to justify and unpack.

**Is there one world? → Yes. We all face the same one world.** Objective monism: regardless of individual perspective, every intelligent agent — human or machine — is anchored to exactly the same single physical reality.

**What is not existing? → Virtual things.** Purely digital constructs, simulations, and unmaterialized concepts lack primary, independent physical existence — they are secondary representations riding on top of the one baseline physical reality, not a parallel reality of their own.

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### Slide 10: World Dynamics — Change

**What is a Change? → Basically the movement of something (change of location).** All change — however different it looks on the surface (chemical, biological, cognitive) — reduces, at bottom, to spatial or structural relocation.

**What Changes? → Objects and Minds.** Only two categories of things experience change: physical objects and (virtual) minds.

**Who does the Changes? → Nature and Humans.** Change is driven either by Nature (spontaneous biological or physical processes) or by Humans (intentional, artificial action) — later expanded in Slide 11 to include a third agent, **AI/“Others.”**

**Are there Other Activities than Movement? → No.** A deliberately radical materialist/functionalist stance: even something as abstract as thinking or learning is, underneath, the functional movement of signals or structures within a network — whether biological neurons or digital logic gates.

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## Slide 12: World — Exercise

**Try to describe a change in the physical world with no movement happening.**

Per the deck’s own framing, this is an intentional trap: any candidate answer (a color fading, a temperature rising, a light dimming) turns out, on inspection, to be movement at a smaller scale — thermal change is atomic/molecular vibration, a color shift is electrons changing energy states, decay is molecular reconfiguration. Consistent with Slide 10’s axiom, there is in fact no valid counterexample: every apparent non-movement change resolves into movement once you look closely enough.

**What moves, when a brain or mind is changed or changes?**

At the physical substrate level: synaptic firing, ion-channel movement, and neurotransmitter diffusion in biological neurons — or, in a machine analog, the movement of electrons through logic gates and the numerical adjustment (“movement” in weight-space) of a neural network’s parameters. The mind is classified as a “mental” entity, but any actual change in it is still, physically, a relocation at the neuronal or electronic level — direct confirmation of Slide 10’s “no activities other than movement” claim.

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## Slide 16: Principles of Entity

**What Characterises an Entity in Principle? → It is delimitable.** The one universal trait of any entity — physical or mental — is that it can be isolated or separated from its surroundings.

**What is a Physical Entity? → A conglomerate of matter and waves.** A physical entity is structurally defined as a packed collection of physical properties.

**What are Delimiters? → e.g. surfaces, different perceptions.** A physical entity’s boundaries are set either by material interfaces (a literal surface) or by where a perceiving mind’s senses register a discontinuity (an edge, a texture shift, a color change) — meaning boundaries are partly observer-dependent, not purely objective facts about matter.

**What is a Mental Entity? → A representation.** A mental entity has no physical surface; it is bounded entirely by its conceptual role as an internal mapping of something.

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## Slide 18: Workshop — Differentiation

**Develop basic categories for initial distinctions between entities.**

The module’s own answer follows on Slide 20: physical entities are sorted along **Mobility** (mobile/immobile), **World Substance** (physical/mental), **Self-acting** (dead/alive), and **Intelligence** (thinker/non-thinker); mental entities (information) are sorted along **Origin** (derived by oneself/learned), **Validity** (verified/unverified), and **Access Control** (confidential/non-confidential). These seven axes are exactly the “basic categories” the workshop exercise is meant to lead students to discover on their own.

**Which categories are most important for us to master our daily life?**

Practically, the survival-critical ones are **Mobility** (is it something that can move toward or away from me — threat or tool?), **Self-acting/Alive-Dead** (is it an agent I must negotiate

with, or an inert object I can simply use?), and **Intelligence/Thinker-vs-Non-Thinker** (can it plan against me, or is it purely reactive?). These let an agent — biological or artificial — quickly triage hazards, tools, and other agents, which matters more for everyday navigation than finer philosophical distinctions.

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### Slide 23: Workshop — Representation

**What different kind of representations do you know?**

Biological neural/synaptic patterns, human language and symbols, mathematical and logical notation, digital database schemas, vector embeddings, pixel/sensor maps, mental imagery, and memories — plus, per Slide 21’s **Notion Hierarchy**, purely conceptual/abstract representations running from **Representation** → **Concept** → **Abstraction** → **Instantiation**.

**Are representations real or virtual?**

Per Slide 25, representations are virtual/mental in content — active internal simulations or copies, not the physical entity itself — yet per Slide 24’s diagram, they are always materialized by and physically tethered to a real substrate (a biological brain or digital hardware). So a representation is virtual in what it represents but physical in how it exists — an early instance of the module’s closing “Paradox and Dualism” (Slide 38): both mental and physical at once.

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### Slide 25: Representation Fundamentals

**What is a Representation?** → **An active simulation/copy of an existing entity.**

Not a passive snapshot — a live internal simulation continuously running inside the mind’s (or machine’s) processor.

**Can the Mind perceive the entity directly?** → **No. Only through the Representation.** A “Cognitive Wall” separates any mind from the external entity itself; all interaction happens through the internal model, never the raw thing — a modern restatement of the Kantian phenomena/noumena distinction.

**Can a Representation be changed with no consequence for the Entity?** → **Yes.**

Because the representation is a decoupled internal sandbox, an agent can modify, damage, or replay it internally with zero real-world effect on the actual entity — this is exactly what makes simulation, planning, and “what-if” reasoning possible and safe, both for brains and for AI systems running predictive scenarios before acting.

**Can a Representation be identical to the Entity?** → **No.** A representation can never fully match an entity’s true, infinite molecular/quantum complexity — it is always a compressed, lossy approximation, as the Slide 26 pipeline makes explicit (Real Entity → Sensors → Signal Processing → Internal Representation, with information filtered and transformed at every stage).

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### Slide 29: Primary Reality Fundamentals

**What is the Primary Reality?** → **What we face together (Physical World).** The shared, common environment that every agent interacts with face to face.

**Can we change the Primary Reality?** → **Yes. Directly via Physical Action.** You cannot think a change into existence in the physical world — you must act on it.

**Do we have the same Representation of it?** → **No. Each mind has its own.** Despite facing an identical shared physical world, every agent's internal model diverges, shaped by its own senses, position, and limitations.

**Is there an Objective Truth in Primary Reality?** → **Yes.** Underneath the diversity of subjective representations, there remains one singular, verifiable physical ground truth — the standard against which any agent's (including an AI's) internal model must ultimately be validated.

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### **Slide 32 (first instance): n-ary Reality Fundamentals — The Symbolic World**

**What is an n-ary Reality?** → **Shared Representations (e.g., Languages, Rules, Money).** Abstract systems with zero intrinsic physical value, whose entire power comes from collective mental backing.

**Can we change the n-ary Reality?** → **Yes. Indirectly via Communication/Agreement.** Unlike the physical world, you change it by persuading other minds to update their shared model — legislating a new law, renegotiating a currency's value, shifting a language's usage.

**Is there an Objective Truth in n-ary Reality?** → **No. Only Shared Agreements.** There is no physical measurement proving a banknote is “worth” a given amount — its validity is entirely social and consensus-based.

**Can it collapse?** → **Yes. If the agreement stops.** Because these systems are anchored purely to continuous collective belief rather than physical mass, losing that shared agreement makes the system instantly cease to exist — hyperinflation collapsing a currency, or a legal system losing legitimacy, are real-world instances of this.

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### **Slide 32 (second instance): n-ary Reality Fundamentals — Secondary and Tertiary Reality**

**What is Secondary Reality?** → **Representation of the Primary Reality of Another Individual.** My best guess at what you, specifically, perceive and believe about the shared physical world.

**What is n-ary Reality?** → **Representation of a (n-1)-ary Reality of Another Individual.** The general recursive rule: each additional layer models one more individual's model of the layer beneath it.

**What is Tertiary Reality?** → **Representation of a Secondary Reality of Another Individual.** A nested, two-levels-deep simulation — e.g., what I think you think a third person believes.

**How Many Realities Can One Individual have?** → **Depends on the Amount of his or her Known Individuals.** The number of secondary/tertiary/n-ary realities one mind must track scales with the size of their social network, not with raw computational power —

which is why modeling large societies is combinatorially explosive for both human minds and AI systems.

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### Slide 36: Workshop — Reality Description

#### Describe the discourse world in your own words.

A “discourse world” is a deliberately narrow, bounded slice of Primary Reality, scoped to a specific place and time interval — the deck’s own example is lecture room K.0.11 on 27 March 2026, from 13:30 to the present moment. It is small enough to be exhaustively described, yet still contains the full richness of entities, relationships, and change that the broader framework (World, Entity, Differentiation, Representation) applies to.

#### Describe the discourse world using a formal modelling technique or language.

This calls for translating the natural-language description into a structured formalism: an entity-relationship model or ontology listing every entity (people, furniture, devices) with properties drawn from Slide 20’s categories (mobile/immobile, physical/mental, dead/alive, thinker/non-thinker), their part-whole and is-a hierarchies (Slide 21), and time-stamped state changes (per Slide 10’s “change = movement” axiom) — equivalently, a logic-based or object-oriented class/instance model with explicit relations between entities.

#### What is the challenge of modelling discourse worlds?

Even a tiny, well-defined slice of reality hides enormous complexity: every entity needs delimiters (Slide 16), its representation is necessarily lossy (Slides 25–26), and multiple agents in the room may hold divergent Secondary Realities of the same space and people (Slides 29, 32). Formal models (database schemas, OOP, ML) tend to flatten all of this into isolated, static instances, losing the observer-dependent, constantly changing, richly interconnected nature of even a mundane scene — the exact gap between intuitive human understanding and mechanical representation that Slide 4’s Question 3 raised at the start of the module.

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### Slide 38: The Paradox and Dualism Explained

#### “Paradox?” — how can mental entities be part of the physical world?

Via a simple logical chain: (a) mental entities are part of the mental world; (b) the mental world takes place in brains; (c) brains are part of the real, physical world; (d) **therefore, mental entities are part of the physical world.** The paradox is that something felt as purely internal and subjective — a thought — turns out, by this chain, to unavoidably also be a physical phenomenon.

#### “Dualisms?” — how is this paradox resolved?

Not by choosing physicalism or idealism exclusively, but by accepting a **complementary dual-aspect dualism**: thoughts are simultaneously **PHYSICAL and MENTAL** — two inseparable aspects of one and the same thing — and minds simultaneously belong to **oneself** (subjective, private, first-person) **and the world** (objective, public, third-person, since their substrate, the brain, is itself a public physical object). For AGI engineering, this is liberating: one does not need to “solve” consciousness to build a thinking machine — only build a system whose physical

substrate (silicon and electrons) simultaneously manifests genuine functional/mental properties. Both aspects can be true at once, with no contradiction to reconcile.

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## Session D — Time and Space

### Slide 2: Repetition — “Reality” (Review of Module C)

#### What is reality?

Per Module C (see *Session C*): reality is “what surrounds us with ourselves included” — the totality of the physical (matter/waves) and mental (minds/thoughts/representations) domains, all contained within one shared World.

#### Which kind of realities can be differentiated?

Per Module C’s layered ontology: **Primary Reality** (the one shared physical world, with objective truth), **Secondary Reality** (one mind’s representation of another individual’s Primary Reality), **Tertiary Reality** (a representation of someone else’s Secondary Reality — a “belief about a belief”), and the general recursive **n-ary Reality** (a representation of an (n-1)-ary reality), which includes purely symbolic/social constructs like language, law, and money that hold no objective truth, only shared agreement.

#### Who defines the subjective reality and out of what?

Each individual processing mind defines its own subjective reality. Per Module B’s interaction loop and Module C’s Representation architecture, it does so out of: raw sensory **impressions** captured through perception, filtered and transformed through signal processing, and reconstructed as an internal, necessarily lossy **Representation** — an active simulation that can never be identical to the entity it maps, and which the mind can never bypass to perceive the entity directly (Module C’s “Cognitive Wall” / Sandbox Principle).

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### Slide 3: Workshop — Time and Space (7 brainstorming prompts)

#### 1. What is time? Define!

Per the module’s own formal answer (Slides 5 and 7): time is not a physical entity — it is a **Mental Concept**, used to put movements in order. Its precise axiom (Slide 7): “*Time is the experience that one entity is always perceived only once*” — a cognitive mechanism enforcing that every observed configuration of matter/state is a unique, unrepeatable slot in a sequence, even if the same coordinates are revisited later.

#### 2. What is space? Define!

Per Slides 16 and 18: space is likewise a **Mental Concept**, not a physical entity itself, but one that both physical and (normally) mental entities are indexed by. Its formal axiom (Slide 18): “*Space is the experience that two entities are never at one single position and one entity is not changing its size with no reason*” — i.e., an exclusion principle (no two entities share one coordinate) plus a volumetric-stability principle (an entity’s size doesn’t randomly change without cause).

#### 3. What are the consequences of a world without time?

Per Slide 6’s counterfactual cascade: without time, change/movement would happen with zero delay (an instantaneous paradox — an object exists at start, path, and destination simultaneously); entities would lose any dedicated, stable location; without stable locations or discrete states, entity boundaries dissolve entirely (“No Entities?”); and without discrete entities, the world container itself stops functioning (“No World?”). Time is therefore a **guarantor of**

**entities** — it is what keeps events from all happening at once and keeps the environment parseable.

#### 4. What are the consequences of a world without space?

Per Slide 17's parallel counterfactual: without space, movement would face no restrictions or barriers (matter could interpenetrate instantaneously and infinitely); entities would lose any dedicated, stable size or volumetric footprint; without stable size or separation, entity boundaries dissolve ("No Entities?"); and without distinct entities, the container again collapses ("No World?"). Space is therefore the structural **guarantor of delimitability** — what keeps entities bounded, distinct, and separately identifiable.

#### 5. What medium connects time and space together?

**Motion.** Per Slide 25, the module's climactic synthesis: "Motion welds Time and Space together" — a complete description of motion needs only the temporal dimension (when/how long, measured in seconds) and the spatial dimension (where from/where to, measured in meters); their combination yields the derived quantity **Speed** (meters per second). Time and space are not independent, isolated frameworks — they are two complementary mental tools an observer uses to describe and measure the single phenomenon of physical displacement.

#### 6. What entity nature has time: physical or mental?

**Mental.** Per Slide 5: "Is Time a Physical Entity? → No." Time does not exist as tangible stuff out in the objective world; it is a cognitive framework a mind constructs internally, bound to (but distinct from) the physical constraint that change can never happen instantaneously (per the speed of light and Special Relativity).

#### 7. What entity nature has space: physical or mental?

**Mental**, but with a caveat: per Slide 16, space is classified as a **Mental Concept**, yet it refers to *both* physical and mental entities — physical displacement strictly requires space ("Walking through Walls does not Work until now"), and mental entities also normally need space, since thinking, simulating ideas, and organizing knowledge naturally use spatial metaphors, data structures, and virtual topologies (e.g., embeddings, knowledge graphs).

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### Slide 5: Time Fundamentals

**Does Change/Movement Always Need Time? → Yes. Think of the Speed of Light  $c$  and the Special Theory of Relativity.** Physical change or movement can never happen instantaneously — it is bound by a cosmic speed limit, meaning time is a non-negotiable requirement for any change to occur.

**What is Time? → A Mental Concept.** Time is not something sensors capture directly like a photon or sound wave; it's a data-structuring system a mind builds to make sense of change.

**Is Time a Physical Entity? → No.** Confirmed directly, reinforcing that time is constructed cognitively rather than detected physically.

**What is Time Used for? → To Put Movements in Order.** Its sole operational purpose is to sort, sequence, and chronologically order movements/events — enabling causal-chain mapping and sequential action planning.

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## Slide 6: A World Without Time (answers Slide 3, Question 3, in detail)

- **Change/Movement** → without delay (an instantaneous paradox: velocity approaches infinity)
- **Location** → no dedicated location for any one entity (stable coordinates become impossible)
- **Entities** → “No Entities?” — boundaries collapse once states blur together instantly
- **World** → “No World?” — the environmental container itself ceases to function once no discrete entities remain

This cascade is the deck’s proof that time functions as a **Guarantor of Entities** (the module’s own subtitle): remove it, and the world’s structural integrity collapses entirely.

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## Slide 8: Workshop — Time Logic (3 brainstorming prompts)

**Where do we as humans know from, what is earlier and what is later?**

Per the Relative Logic model formalized on Slide 10: humans determine “earlier vs. later” through memory-state comparison rather than an external clock — if a memory of event A already exists while a new impression B is currently being processed, the mind logically deduces A happened earlier. It’s a simple causal/sequential counter built from stored versus incoming experience, not a numerical timestamp.

**What do we need to bring the time points of events/movements into an order?**

At minimum, a way to distinguish “already experienced” from “currently experienced” (an impression/event counter, per Relative Logic); for coordinating across separate minds, threads, or distributed devices, this isn’t enough — you additionally need a shared, synchronized measurement system with a common anchor and fixed distance units, i.e., **Absolute Logic** (a fixed reference point plus clocks providing defined, uniform time distances, per Slides 10–11).

**What are natural time measurements?**

Astronomical cycles: the solar **Year**, subdivided into **Months** (1/12 of a year) and **Weeks** (7 days), and the solar **Day**, subdivided into **Hours** (1/24 of a day) — per Slide 11, these are the natural periodic phenomena (Earth’s orbit and rotation) that historically anchor all absolute calendar systems.

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## Slide 10 / 11 / 12: Relative vs. Absolute Time Logic

**What distinguishes Relative from Absolute time logic, and why do both matter?**

**Relative Logic** is the natural, biological default: no clock required, just a memory-based sequence counter using a floating reference point (“now,” “today”), with past/future indexical vectors (yesterday, tomorrow). It’s ideal for local, single-mind or single-thread causal reasoning, but breaks down the moment multiple independent threads or agents need to compare notes — at that point a shared synchronizing anchor becomes necessary. **Absolute Logic** is the human-invented alternative: a fixed global reference event (e.g., the Birth of Christ, giving BC/AD) plus clock-measured, uniform distances (years, months, hours), so that entirely separate, distributed observers can align their timestamps and compare events without ever needing to have witnessed the same moment directly. Robust AGI/multi-agent architectures need both: relative logic

for fast, local, causal reasoning, and absolute logic for distributed synchronization, network consensus, and database alignment (Slide 10's interpretive note on race conditions).

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### Slide 13: Clocks — The Materialization of Time

**What is a Clock?** → **An Artificial Machine that Materialises Humans' Concept of Time.** A clock does not detect or capture some physical substance called “time” from the environment; it is a state machine built specifically to give physical form to an internally-defined mental concept.

**Could a Human do the Work of a Clock?** → **Yes.** Because the underlying task is a purely logical/counting operation, a human mind can replicate it manually (e.g., counting a steady rhythm or interval).

**What is the Work of a Clock?** → **Counting the Defined Measurement Units.** Its entire function reduces to counting regular, repeatable units (mechanical swings, crystal oscillations, atomic transitions) and incrementing a sequence.

**Is Clock Work Mental Work?** → **Yes.** Since it is fundamentally an exercise in counting, indexing, and updating a sequence chain — a logical process — clockwork is classified as mental work, whether performed by gears, electronics, or a human mind.

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### Slide 14: Workshop — Clock Systems Analysis (three “workers”)

Mapping the analog clock, a human, and an atomic clock through the Work System framework (why / who / with / how / what, from Module A):

- **(a) Analog clock** — *why*: to materialize and visibly display the passage of time for human reference; *who*: the mechanical clock mechanism itself; *with*: a spring or weight-driven escapement, gear train, balance wheel/pendulum, and hands; *how*: counting regular oscillations of the balance wheel or pendulum and converting each count into incremental rotation of the hands; *what*: a continuously updated visual position encoding the current time.
- **(b) Human** — *why*: to estimate elapsed time or order events for coordination and daily activity; *who*: the person themselves; *with*: memory, attention, and an internal biological rhythm (e.g., counting seconds mentally, or using heartbeat/breathing as a rough pulse); *how*: mentally counting fixed subjective intervals or comparing “already experienced” to “currently experienced” states (Relative Logic, Slide 10); *what*: an approximate, subjective estimate of elapsed time or event order, prone to memory drift.
- **(c) Atomic clock** — *why*: to provide an extremely precise, globally synchronizable reference time (e.g., for GPS, telecom networks, scientific timing); *who*: the cesium/rubidium atomic oscillator apparatus; *with*: the hyperfine transition frequency of a cesium-133 atom (defined as exactly 9,192,631,770 oscillations per SI second); *how*: electronically counting these quantum oscillations; *what*: an ultra-precise absolute timestamp output that can be broadcast to synchronize distributed computing nodes worldwide.

All three are functionally/architecturally identical at the logical abstraction layer (per the slide's own conclusion): they all just count defined units — only the physical medium executing the count differs.

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## Slide 15: Workshop — Space (3 brainstorming prompts)

### What are natural measurement units for space?

Body-relative units biological organisms use without formal tools: steps/paces, arm spans, hand widths, field-of-view/sightlines, or the reach of a limb — natural, embodied reference scales that predate standardized metrics like the meter.

### How many dimensions has space?

Per the module's fuller answer on Slide 20: a comprehensive pose model needs **9 dimensions total** — 3 for **Location** (X, Y, Z translation) and 3 for **Orientation** (roll, pitch, yaw rotation), beyond the basic 3-dimensional (X, Y, Z) view of space that ordinary physics uses for position alone.

### Imagine if bodies would not need space. What would happen?

If physical entities had no spatial footprint or displacement requirement, all matter in the universe could occupy the exact same coordinate at the exact same moment — collapsing the entire physical world container into a single, infinite-density point. This is exactly the counterfactual cascade Slide 17 spells out in detail (see below).

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## Slide 16: Space Fundamentals

**Does Change/Movement Always Need Space? → Yes. Walking through Walls does not Work until now.** Physical displacement or structural alteration strictly requires space to move through; solid boundaries cannot be crossed without spatial translation.

**What is Space? → A Mental Concept.** Mirroring time, space is not a physical object itself but an internal cognitive framework.

**What does Space Refer to? → To Physical or Mental Entities.** This mental concept is applied to index both tangible, physical objects and abstract, mental ones.

**Do Mental Entities Need Space? → Normally Yes.** Even purely mental activity — thinking, simulating ideas, organizing software schemas — is naturally structured using spatial metaphors, data structures, or virtual topologies (e.g., vector embeddings, knowledge graphs).

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## Slide 17: A World Without Space (answers Slide 15, Question 3, in detail)

- **Change/Movement** → without restrictions (matter could interpenetrate instantaneously and infinitely)
- **Size** → no dedicated size for any one entity (no spatial grid to measure a stable, unique scale)
- **Entities** → “No Entities?” — individual boundaries dissolve once size and separation are gone
- **World** → “No World?” — without distinct entities to populate it, the container breaks down

This confirms space’s role as the module subtitle names it: a **Guarantor for Delimitability** — the structural mechanism that keeps entities bounded, distinct, and separately identifiable.

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### Slide 19: Workshop — Space Logic (formalizing a real position)

**Describe in your own words unambiguously the position of the two big screens in your discourse world “Intuition of AI.” / Describe the position using formal variables and values. / Extend your UML diagram by the description of the positions of the two big screens.**

This is a three-layer translation exercise building directly on Module C’s “discourse world” exercise (K.0.11, Slide 36): (1) **natural-language description** — e.g., “Screen 1 is mounted on the front wall, centered, about 2 meters above the floor, facing the seating area; Screen 2 is mounted to its right at the same height”; (2) **formal variables** — assign each screen a coordinate and orientation, e.g., `Screen1.location = (x1, y1, z1)`, `Screen1.orientation = (roll1, pitch1, yaw1)`, using the room’s corner or the lecturer’s position as the **Location Reference Entity** (per Slide 22’s 6-parameter relative model); (3) **UML extension** — add `location: Vector3` and `orientation: Vector3` attributes to a `Screen` class, associated with the `Room/DiscourseWorld` class from the Module C exercise, so the spatial description becomes part of the same formal object model rather than a separate afterthought.

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### Slide 20: Space Logic — Dimensionality and Units

**How Many Dimensions? → 9 total: 3 Location + 3 Orientation.** A full “pose” of an entity requires 3 degrees of freedom for position (X, Y, Z) plus 3 degrees of freedom for rotation (roll, pitch, yaw) — going beyond the naive 3-dimensional view of “space” as position alone. (The slide’s own bullet notation contains a minor transcription inconsistency, flagged in the notes, but the clean  $3+3 = 6$  structure is what full pose-estimation systems actually use; the deck’s “9” figure appears to be a slide-text artifact.)

**What for? → Organize positions (in work systems).** Dimensionality exists to let a work system consistently locate and orient its participants and objects relative to one another.

**Typical Unit? → Meter: 1/299,792,458 of the distance light travels in a vacuum in 1 second.** Just like the astronomical/atomic basis for the second, the modern meter is itself defined via a physical constant (the speed of light) — showing that space’s fundamental unit cannot be defined without already relying on a precise temporal calculation.

**Needed for? → Space Logic.** These units and dimensions are the necessary substrate on which all subsequent spatial-logic operations (relative or absolute positioning) are built.

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### Slide 21 / 22: Relative vs. Absolute Space Logic

**What distinguishes Relative from Absolute space logic, and why do both matter?**

**Relative Logic** mirrors biological spatial cognition: no global grid, just nested “is part of”/“part-whole” hierarchies (the keyboard is part of the computer, which is part of the room), with position and orientation computed relative to a known reference point (Slide

22's 6-parameter model: a Reference Entity, its two reference points/"face," the target's reference point, its Location, its Orientation Point, and its Orientation — yielding intuitive directions like *left/right, up/down, in front/behind*). **Absolute Logic** is the human-engineered alternative: a synchronized global grid using latitude, longitude, and altitude, anchored to fixed compass directions, so any entity can be tagged with unique coordinates valid across distributed, non-local observers. Relative logic is efficient for local manipulation and human-robot interaction ("hand me the wrench behind you"); absolute logic is mandatory for global path planning, fleet telemetry, and cross-node spatial mapping.

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## Slide 25: The Synthesis of Motion

**What is the medium/mechanism that ties time and space together, and what does describing it fully require?**

**Motion.** Time and Space are not independent, free-standing realities — they are complementary mental tools an observer builds specifically to measure and describe physical displacement. A complete description of motion needs only two dimensions: the **temporal** (*when? how long?*, measured in seconds) and the **spatial** (*where from? where to?*, measured in meters); when a mind tracks an entity's changing state across both simultaneously, they weld together into the derived kinematic quantity **Speed** (meters per second, tracking *path & rotation*). This applies equally to tangible objects moving through physical space and to abstract data states shifting within an internal mental sandbox — the same underlying space-time logic indexes both physical and mental entities.

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## Slide 26: Conclusion

**Are time and space separable? → No. Inseparability.** They are the fundamental, jointly-necessary measures for the motion of physical entities.

**Are humans experts in time and space logic? → Not really — but they are experts in communicating about it.** Human memory is fallible and imprecise at tracking exact durations or positions, yet humans are remarkably skilled at *communicating* temporal and spatial information to each other (natural language relative terms like "tomorrow" or "behind you," per Slides 8/12/21).

**Do current formal languages support native temporal/spatial logic? → No — this is a gap, addressed by frameworks like GREBTL.** Standard programming/formal languages have no built-in temporal or spatial logic; the presenter's own proprietary framework (GREBTL, referenced under "Advanced Information Modelling") is offered as a step toward closing this gap.

**What does this mean for building true AGI?** Any system aiming for genuine intelligence must have **intrinsic** temporal and spatial logic built into its core architecture — the way humans have it — rather than treating time and space as external metadata bolted onto an otherwise time/space-blind system. This is presented as a major open opportunity in software engineering and AI development, not a solved problem.

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## Session E — Information

### Slide 2: Repetition — “Time and Space” (Review of Module D)

#### What is used to define the measurement unit of time?

The **second**, ultimately anchored (per Module D) to natural astronomical cycles — the solar year subdivided into months (1/12) and the solar day subdivided into hours (1/24) — and, for atomic-clock precision, defined via a fixed number of cesium-133 hyperfine oscillations. See *Session D*.

#### What is used to define the measurement unit of space?

The **meter**, defined (per Module D, Slide 20) as 1/299,792,458 of the distance light travels in a vacuum in one second — meaning the fundamental spatial unit is itself derived from a precise temporal calculation, illustrating how deeply time and space are already interlinked at the definitional level.

#### Is time/space relative or absolute? Why?

Both, depending on the use case (per Module D, Slides 10, 11, 12, 21): **Relative** logic (no clock/grid needed, just memory-based sequencing or nested “is part of” positioning from a floating reference point like “now” or “me”) is the natural, biological default and is efficient for local, single-agent reasoning. **Absolute** logic (a fixed global anchor plus synchronized, uniform units — calendars with a reference epoch, or latitude/longitude/altitude grids) is a human invention required whenever multiple independent agents or distributed systems need to compare and synchronize events or positions without having witnessed the same moment or place directly.

#### Even if time and space are mental entities, how can we get an idea of time and space by impressions or perceptions from the physical world?

Because time and space, while themselves mental concepts (not physical stuff), are the cognitive filters a mind applies to raw sensory impressions of physical change. Physical movement/change (per Module C/D) always requires time and space to occur, so when an impressor (sense organ or sensor) registers a physical entity’s displacement, the mind indexes that impression using its internal time and space frameworks — the impressions come from real physical change, but the mental *dimensions* used to organize them (order, distance, duration) are constructed internally, not detected as separate physical substances.

#### What binds time and space inseparable together?

**Motion** (per Module D, Slide 25) — a complete description of motion only needs the temporal dimension (when/how long) and the spatial dimension (where from/where to); tracking an entity’s change of state welds the two together into the derived quantity Speed.

#### How would you explain, that space is not existent?

Space is not a physical substance you can isolate, bottle, or detect directly with a sensor — no instrument measures “space” the way a thermometer measures temperature. What is physically real is only matter and waves and their relative displacement; “space” is the mental framework (Module D, Slide 16: “What is Space? → A Mental Concept”) a mind builds to describe and enforce that two entities cannot occupy the same coordinate and that an entity’s size doesn’t randomly change (Slide 18’s axiom). Remove the observing mind, and there is no separate “space” left over — only matter in relation to other matter.

## How would you explain, that time is only a mental phenomenon?

Similarly, no sensor captures “time” as a physical particle or wave the way it captures light or sound (Module D, Slide 5: “Is Time a Physical Entity? → No.”). What is physically real is only that change/movement cannot happen instantaneously (bounded by the speed of light). Time is the mental bookkeeping device — “the experience that one entity is always perceived only once” (Slide 7’s axiom) — that a mind constructs specifically to sequence and order those physical changes; it is a cognitive tool for organizing perception, not a substance existing independently “out there.”

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## Slide 3: Workshop “Information” (5 brainstorming prompts)

### Where does Information exist?

Strictly **inside a processing mind** — a brain or “thought machine” (per Slide 17). Information does not float freely in the physical world; per the module’s central thesis (elaborated across Slides 7, 17, 22), physical reality contains only matter and waves — no time, no space, and no information natively exist “out there.” Information only comes into existence once a mind processes raw physical impressions and represents them internally.

### What is Information? Define!

Per Slide 17’s formal thesis: Information is a **mental entity in the mental world** — always part of a brain or thought machine (itself physically embedded in the world, per body-mind dualism), always **representing** some other entity (a pointer, not the thing itself), and scalable across three granularities (atomic, elementary, complex — see Slide 24). Per Slide 22’s rapid-fire summary: Information = Mental Entity = Representation, and it is **always** mental — never physical.

### What is an Information Theory?

Per Slide 15’s workshop framing: a theory’s task is to provide a logically consistent model that can **explain** past phenomena and **predict** future ones. A proper “Information Theory,” then, must give a measurable, calculable way to predict how information behaves, degrades, and can be reliably transmitted — which is exactly what Claude Shannon’s mathematical theory of communication provides, and (per Slide 16) it turns out to sit at the hub of a much wider “flower” of disciplines: communication theory, probability, statistics, mathematics, economics, computer science, and physics all share its core mathematics (compression limits, entropy, Kolmogorov complexity, thermodynamics).

### What is the Difference between Information and Knowledge?

Per Slides 27, 29, and 31: **Information** is a passive representation of the world — a map, useful for building an internal model, but inert on its own (e.g., knowing EU population statistics is just information). **Knowledge** is what Information becomes once it is connected to other information networks in a way that enables **purposeful acting** — the ability to actually *use* the map to navigate. Formally: Information is the base for reasonable thinking; Knowledge is needed for purposeful acting; and even thinking itself counts as “mental acting” (Slide 31).

### Where is the data?

This question deliberately sets up the Data/Information/Knowledge distinction that Slide 8 makes explicit: raw **data** (magnetic states on a hard drive, binary in a computer, tick counts in

a clock, coordinates from a GPS) sits in physical storage media — but it is *not yet* information. Data only becomes Information once an observing mind reads, parses, and assigns meaning to it within its own subjective reality (Slide 8’s repeated “Information?” question mark next to every artificial system’s output underscores that machines by default only produce Data, Time, or Dimensions — not Information — until a mind interprets them).

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### Slide 8: Do Machines Produce Information?

#### Do artificial systems (computers, clocks, GPS) natively produce Information?

No — per Slide 8’s deliberately repeated, questioned “Information?” label. A clock produces a tick count; a computer produces binary data; a GPS produces spatial coordinates. None of these become **Information** until an observing mind (a “Homo Sapiens Brain,” per the slide’s comparison) looks at them, parses them, and assigns them meaning within a subjective reality. Only the biological brain is shown natively generating Information, Knowledge, and Thoughts — every artificial system’s output remains mere Data until a mind processes it.

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### Slide 11: Workshop “Complexity” (5 brainstorming prompts)

#### Which impressions do we store in our memory and which not?

Only a small, filtered subset — those judged relevant to survival, goals, or attention at the moment of perception. The overwhelming majority of raw sensory impressions are discarded immediately, since a system cannot physically store the infinite, chaotic stream the physical world generates.

#### What or who decides, about the impressions to be stored?

The mind’s own attention and relevance-filtering mechanisms — driven by factors like survival needs, active goals, novelty, and emotional salience — act as the gatekeeper. This is the same role that engineered systems fill with explicit filtering rules (e.g., an autonomous vehicle’s object-detection thresholds deciding which LIDAR points to keep).

#### What do you think, how impressions are transformed into information?

Per the module’s later formal answer (Slide 19): raw impressions are transformed into information through internal **Processing** — the mind takes an incoming impression, runs it through its interpretive/cognitive machinery (perception, recognition, pattern-matching, or reasoning), and outputs a structured mental entity (Information) that represents the original stimulus, ready to be expressed or acted upon.

#### How is our mind organising the complexity of all the perceived impressions?

Via the “Simplifier” tools detailed in Slide 13: **Abstraction** (finding equivalence relations and keeping only the essential structure), **Classification** (assigning entities to rigid, formal groups), **Categorisation** (assigning entities to broader, fuzzier mental groups), and **Reduction** (simply omitting/discarding data without guaranteeing the essential parts survive).

#### What constructions is our mind using to simplify or reduce complexity?

The same four tools — Abstraction, Classification, Categorisation, and Reduction (Slide 13) — plus, at a higher level, the **Concept Schema** mechanism introduced later in the module (Slide

34): pre-learned templates (like “father → has children”) that let the mind instantly slot new entities into an existing structure rather than processing everything from scratch.

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## Slide 12: Complexity — Fundamentals

**What is Causing Complexity? → Big Systems with a Lot of Entities Interacting and Moving in Time and Space.** Complexity arises from scale and interaction: as more entities move and interact across time and space, the number of possible states grows explosively.

**What is Quantity/Space Complexity? → When Space is Crucial to Solve a Problem.** A problem has spatial complexity when solving it requires tracking large numbers of coordinates or volume — directly analogous to memory/RAM usage in computer science.

**What is Time Complexity? → When Time is Crucial to Make a System Work or Solve a Problem.** A problem has temporal complexity when solving it requires tracking long chronological chains, delays, or strict deadlines — directly analogous to processing time/CPU cycles.

**Where is the Complexity? → Inside Brains.** The physical world itself is just matter moving around; it doesn’t “know” it is complex. Complexity is entirely the cognitive burden placed on a mind or processor trying to model, track, and predict all those interacting entities — meaning complexity is an information-processing problem, not an environmental one, and is solved by building better internal simplifying algorithms rather than changing the world.

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## Slide 13: Complexity — Methods of Handling

**Abstraction? → Using Equivalence Relations, Omitting Details with Focus on the Essential.** The most rigorous simplifier: it finds shared structural/mathematical properties across different entities and strips away noise until only the core mechanism remains (e.g., a physics simulation ignoring friction to isolate baseline gravity).

**Classification? → Assign Entities to Groups (narrow/more formal).** A rigid, hard-coded sorting mechanism using strict, predefined rules (e.g., biological taxonomy, database schemas).

**Categorisation? → Assign Entities to Groups (broad/more mental).** A softer, fuzzier sorting mechanism based on subjective utility rather than strict rules (e.g., clustering algorithms, “things you take on a picnic”).

**Reduction? → Omitting Details without Strict Focus on the Essential.** The bluntest tool: it deletes or ignores data to save space, but — unlike abstraction — offers no guarantee that what remains is actually the important part (lossy compression; risky if applied to safety-critical parameters).

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## Slide 15: Workshop “Theory”

**What is the task of a theory, and what should a theory for information provide?**

A theory's task is to provide a logically consistent model that can **explain** past phenomena and **predict** future ones — it isn't just an arbitrary list of facts. A valid “theory for information,” specifically, must therefore provide a measurable, calculable way to predict how information behaves, degrades, and can be reliably transmitted — the exact requirement Shannon's Information Theory was built to satisfy, and which underpins modern data compression and error-correction engineering.

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## Slide 17: Information Theory — Thesis

### What is information?

Per the five-part thesis: (1) Information is a mental entity in the mental world; (2) it is always part of a brain or thought machine, which is itself part of the physical world (body-mind dualism); (3) it always represents other entities — it is fundamentally a pointer, not the thing itself; (4) it has three granularities — atomic, elementary, complex (see Slide 24); (5) processing (logic) and persisting (memory) of information is the task of a brain.

### What is an entity?

An entity is **everything that is delimitable** — “No delimitability, no entity.” This single criterion applies equally in the physical world (a rock has physical edges) and the mental world (a specific thought or variable must have a defined boundary to be processed at all).

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## Slide 19: Information Theory — How is Information Emerging?

### How is information emerging?

Through three distinct pathways: **(1) Perception (exogenous)** — either explicit/intentional (active learning, e.g., studying a book) or implicit/unintentional (passive observation, e.g., an unexpected loud noise); **(2) Thinking (endogenous)** — either explicit, step-by-step conscious logic, implicit via educated natural neural networks (biological intuition/“gut feeling”), or implicit via trained artificial neural networks (deep-learning pattern recognition — placed on the same footing as human intuition); **(3) Sharing** — either asynchronous (time-delayed, e.g., books, recordings) or synchronous (real-time, e.g., speech, phone calls). These feed a continuous cycle: **Impression → Processing → Expression**, where one agent's Expression becomes another agent's new Impression.

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## Slide 20: Information Theory — Utility and Communication

### What is information needed for?

To build a **representation of the world** an agent lives in, so that it can solve daily problems and survive — information exists to enable action, not for its own sake.

**Is Communication the same as Information? → No — Communication is not information.**

Information is a purely mental entity, existing only inside a brain/CPU; it can never literally leave that mind. What “communication” actually does is **materialize** a physical copy of an

internal mental state — sound waves, radio signals, electrical pulses — onto a media channel. That physical signal crosses to a receiver’s sensors, which then generate an entirely **new** piece of information inside the receiver’s own separate mental world. In short: information leaves the mental world only as a materialized physical copy, and re-enters the mental world (in the receiver) as freshly-generated information — never as a direct transfer of the original mental entity itself.

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## Slide 22: Information — Fundamentals

**What is Information?** → **Mental Entity.** **What is a Mental Entity?** → **A Representation.** **Is Information always Mental?** → **Yes.** **Where is Information?** → **Inside Brains.**

Together, these four rapid-fire answers cement the module’s core thesis: information never exists in a vacuum, a wire, or a piece of paper — it exists strictly as an internal representation inside a processing engine (biological or artificial). Storing a file on a hard drive is storing physical magnetic polarities, not “information” — it only becomes information again once a CPU reads, parses, and represents those polarities internally.

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## Slide 25: Information — Basic Information Characteristics

**Entity Essence?** → **Understanding of an Entity with its Complexity.** The deep, network-level understanding of what a thing actually is and how it relates to everything else (its full graph, per Slide 24).

**Entity Type?** → **Category or class.** The broader category an entity belongs to — directly analogous to a `Class` in object-oriented programming.

**Entity Identifier?** → **Unique name.** A strict, unambiguous key used to track the entity without confusion (e.g., a UUID or database primary key) — necessary to distinguish, say, two different people who share a common name.

**Entity Name?** → **Common Identifier.** The everyday, human-readable label used to refer to the entity (e.g., “Eiffel Tower”), distinct from its strict unique identifier.

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## Slide 28: Workshop “Knowledge” (6 brainstorming prompts)

**What was considered as being intelligent?**

Per Module B’s Duden-based definition (see *Session B*): the ability to think abstractly and reasonably and to derive purposeful action from that — a standard this module now re-applies to distinguish mere information-possession from genuine intelligence.

**Why is just information not sufficient to solve challenges?**

Because information is a passive representation — a map, not a plan. Per Slide 29, knowing the EU’s population figures or a city’s geography is still just inert data storage until the system figures out how to *use* that information to achieve a goal. An LLM holding billions of parameters

(massive information) still cannot solve a problem on its own without an objective, prompt, or action pathway connecting that information to a purpose.

### **What makes an information valuable for solving challenges?**

Its **connectivity to other information networks in a clever way** that enables purposeful action (Slide 29) — value isn’t in the raw fact itself, but in whether the information can be linked to a goal-relevant network (e.g., geographic information linked to a transportation or economic network to plan a trip or an investment).

### **Why do we learn? Why do we want to know more things?**

Because acquiring new information/knowledge is what allows an agent to close the distance between its current model of the world and what it needs to successfully act — learning is in service of purposeful acting (Slide 30’s intelligence definition), not an end in itself.

### **Where do we know from, what we want to learn?**

From detecting **Knowledge Gaps** — per Section 6 of the module (Slides 33–35): when an agent’s existing Concept Schemas have empty, unfilled slots relevant to a current task, that gap is what specifies exactly what needs to be learned next.

### **Where does the motivation come from?**

From the psychological/algorithmic discomfort of an unfilled **Knowledge Gap** (Slide 32’s section header links “Knowledge Gap” directly to “Motivation”) — in AI terms, this is analogous to a reward function or an active objective that flags missing information required to complete a task, driving the system to seek it out (see Slides 34–35 below).

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## **Slide 31: Knowledge — Fundamentals**

**What is the Base of Knowledge? → Information.** Without information (a representation of the world), there is nothing for knowledge to be built from.

**For what is Knowledge the Base? → Reasonable Thinking.** Knowledge — connected, purpose-relevant information networks — is what makes reasoning possible; you cannot reason without a knowledge network to traverse.

**What is Knowledge needed for? → Purposeful Acting.** This is Knowledge’s entire function: enabling an agent to derive and execute goal-directed behavior, not merely to hold data.

**Is Thinking Acting? → Yes. Mental Acting.** Action is not restricted to physical movement — the module explicitly defines “thinking” itself as a valid form of **mental acting**, meaning an AI updating its internal weights, restructuring its own knowledge networks, or running an internal simulation counts as genuine action, even before anything physical occurs.

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## **Slide 33: Workshop “Knowledge Gap”**

### **How can we know about knowledge gaps (consider the information theory)?**

By treating gaps as structural, mathematical voids in a data schema rather than mystical feelings of curiosity. Per the module’s own follow-up answer (Slide 34): an agent that has

learned a **Concept Schema** (an empty template — e.g., “father → has one or more children”) automatically detects a gap whenever it encounters an instance of that schema with unfilled slots (e.g., hearing “Robert is a father” but not how many children he has). The gap is detectable precisely because the schema defines what information *should* be present; a missing slot in that schema is the concrete, structural signal of a knowledge gap.

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### Slide 34: Knowledge Gap — Concept Schemas

#### How does an agent recognize and fill a knowledge gap in practice?

Through a three-step execution loop: (1) upon encountering a new entity (“Robert is a father”), the agent instantly retrieves the matching learned **Concept Schema** (the abstract template “father → child 1 ... child n”); (2) it detects the **Information Gap** — the schema’s child-slots are empty for this specific instance; (3) it is thereby motivated to act — asking a question (“How many children does he have?”) to fill the missing atomic information, completing the network (“Robert → child 1, child 2, child 3”). This is directly analogous to slot-filling in NLP systems (e.g., a flight-booking assistant instantiating a [Destination]/[Date]/[Passenger Name] schema and asking follow-up questions for any slot left empty) — it is how curiosity and motivation can be engineered into an AGI: giving it world templates and compelling it to complete them.

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### Slide 35: Knowledge Gap — Fundamentals

**What is a Knowledge Gap? → The Information we Need to Fulfill a Task.** A gap is only a “gap” relative to a specific task — it’s whatever missing data prevents that task’s completion.

**Why do we Recognise Gaps? → We want to complete incomplete schemas.** The underlying drive is the desire to fill the empty slots in an internal concept template rather than leave it structurally unfinished.

**When do we Recognise Gaps? → While Making Action Plans.** Gaps surface specifically during planning — e.g., realizing “I want to visit Robert, but I don’t have his address” only becomes apparent once you try to construct the concrete plan of action.

**When do we Look for Concept Schemas? → When we Fail in Solving a Problem.** If existing schemas prove inadequate — i.e., the agent fails to solve a problem using its current templates — that failure is what triggers the search for an entirely *new* concept schema to re-model the problem differently, rather than just refilling an existing one.

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## Session F — Action

### Slide 2: Repetition — “Information” (Review of Module E)

#### What is information?

Per Module E (see *Session E*): Information is a **mental entity** — a representation that exists only inside a brain or processing engine, always pointing to (representing) some other entity, and scalable across three granularities (atomic, elementary, complex).

#### Where is information?

**Inside brains** — never in a vacuum, a wire, or a piece of paper. A hard drive stores physical magnetic polarities, not information; it only becomes information again once a processing mind reads and represents it internally.

#### How is information transferred from sender to receiver?

It cannot be transferred directly, because Information never leaves the mental world as itself. Per Module E’s strict distinction (“Communication is not information”): the sender **materializes** a physical copy of their internal mental state into waves (sound, light, electricity) on a media channel; this physical signal crosses to the receiver’s own sensors (impressors), which then generate an entirely **new** piece of information inside the receiver’s separate mental world. Nothing mental is literally copied — only a physical signal is, and the receiving mind must reconstruct meaning from it.

#### Can information be deleted and how?

Yes — by destroying the physical storage medium (the brain tissue or the hardware memory) or by reallocating/overwriting the memory space that held it, which causes the corresponding mental representation to cease to exist. Since information only exists as a mental entity dependent on a physical substrate, destroying or repurposing that substrate is the only way to delete it.

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### Slide 4: Workshop “Action” (4 brainstorming prompts)

**1. What is an action? Define that as precise and as abstract as possible. Think also of the work matrix we used in chapter A.**

Per the module’s own eventual formal answer (Slide 20/22): an action is a **change (manipulation) in the physical or mental world** — it always creates, changes, or deletes an entity, and it is always combined with motion/movement. Tied back to Module A’s Work System matrix (Actor/Who, Goal/Why, Tool/With, Process/How, Object/What): an action is what happens when a **Worker** applies a **Tool** through a **Process** to transform a **Work Object** in pursuit of a **Purpose** — i.e., “work” and “action” are the same underlying phenomenon, described at different levels of abstraction (Module A’s socio-technical framing vs. this module’s cognitive-mechanistic framing).

**2. Who can execute actions? Or what is needed to execute actions?**

Per Slide 6’s three prerequisites, an **Actor** needs: (1) the ability to **move** (an actuator — a hand, a motor, or write-access to a database — something capable of changing a state); (2) the ability to **think** (a brain/processor, because purposeful acting requires anticipation, planning via abstraction, and learning from outcomes); and (3) the ability to **communicate**

(a transmitter, so the actor can move information to other actors and coordinate work) — with an explicit exception carved out for fully isolated “lone wolf” systems.

### 3. Which different types and purposes of actions would you distinguish?

Per Slide 7’s directional taxonomy: actions are either **Outwards** (external) — **Act by body** (physical movement/manipulation) and **Act by speech** (communication/information transfer) — or **Inwards** (internal) — **Act by thinking** (internal mental processing). Per Slide 10, physical acts additionally split by *purpose*: purely **Physical** (moving from A to B), purely **Mental** (moving the body just to change someone else’s mind, e.g., body language), or **Hybrid** (moving physically specifically to acquire or set information, e.g., walking around a corner to see what’s there).

### 4. Do you think, an action is the only connection between the mental and the physical world? Can you find other connections?

The module’s own definitive answer (Slide 21) is **Yes** — **Action is the only connection**, described as the “GLUE” binding the mental world (thoughts, schemas, databases) to the physical world (atoms, geography, bodies), which Body-Mind Dualism otherwise treats as entirely separate domains. Concretely, this glue works bidirectionally: when the physical world acts on the mind (light hitting an eye), it creates an **Impression**; when the mind acts on the physical world (a hand moving a box, or speech triggering another’s action), it creates an **Expression**. No other bridge is offered in the module — perception (Impression) and action (Expression) are treated as the two directions of the single Action/Impression-Expression channel, not as separate, independent connections.

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## Slide 6: Actor — Prerequisites

### What must an entity have to qualify as an “Actor”?

Three prerequisites, all necessary together: it **must be able to move** (an actuator capable of changing a state); it **must be able to think** (because purposeful acting requires anticipation — no purposeful acting without anticipation — and planning means abstracting, plus the capacity to learn from results); and it **must be able to communicate**, except for entirely isolated “lone wolf” systems. An AI with only an LLM and no API access to act on the world fails prerequisite 1; a robot arm that can move but cannot anticipate or plan is a dumb tool, not an actor, failing prerequisite 2 — a true Agentic AI needs all three simultaneously.

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## Slide 12: Workshop “Thinking Act” (4 brainstorming prompts)

### 1. Why is thinking an action? What is thinking doing?

Thinking qualifies as an action because — per Slide 20’s definition of action as any change in the physical *or mental* world — thinking changes the internal mental world even when nothing external is altered. It updates schemas, creates new knowledge networks, or modifies action plans; per Module E (Slide 31), thinking is explicitly classified as “**Mental Acting**.”

### 2. Which “tools” are used while thinking?

The internal processing machinery of the mind: perception and recognition modules (for interpreting incoming impressions), and the core engines of **Solving, Learning, Information,**

and **Knowledge** (per Slide 13’s processing-loop diagram) — biologically, neurons and memory circuits; computationally, logic gates, memory registers, and (for an AI) things like vector databases, RAG pipelines, and decision trees.

### 3. What is changed by thinking?

The agent’s own internal **Information** — its stored representations, schemas, and knowledge networks — is created, updated, or restructured. Since thinking doesn’t act directly on the physical world, whatever it changes must be internal/mental (per Slide 14’s “endogenous loop”: “*change information by external or internal events*”).

### 4. How can we determine a need for action?

Per the module’s later, formal answer (Slides 20 and 22): a need for action is determined **when perceived impressions do not correspond to (do not meet) the agent’s inner values** — i.e., when the agent’s model of the current state of the world diverges from its target/goal state. This mismatch is the trigger for both mental action (updating the model) and, ultimately, physical or speech action to close the gap.

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## Slide 13 / 14: The Thinking Act — Processing Loop and Internal Change

### What is the Thinking Act, formally?

The central processing hub of intelligence, sitting between incoming Impressions and outgoing Expressions: the **processing of impressions, solving challenges, learning, planning, and executing purposeful actions after determining a need for action**. Impressions enter and undergo Perception/Recognition, feed into the core engines (Solving, Learning, Information, Knowledge), and the engines output commands to Motion and Speech, which exit as Expressions.

### Does the Thinking Act always need external input?

No — per Slide 14, a thinking act can be entirely self-contained via an **endogenous loop**: the mind can trigger its own internal events and change its own information with no new data from the physical world (analogous to an AI running self-play or background processing/refactoring entirely in memory, disconnected from any sensor).

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## Slide 16: Workshop “Speech Act” (4 brainstorming prompts)

### 1. How can speaking be an action?

Because speaking satisfies the definition of action (a change in the physical or mental world combined with motion): it physically changes the environment (pushing sound waves through the air) and it changes the mental environment of the listener (altering their internal information state).

### 2. What is changed during a speech?

Two things simultaneously: the **physical world** (sound waves propagate through air, or light/electrical signals if written/digital) and the **listener’s mental world** (their internal representation/information is updated by the received message).

### 3. Which types of sentences do you know?

Per Slide 17's formal taxonomy: **Declarations** (changing another individual's information by stating a fact), **Questions** (getting information from another individual to fill one's own knowledge gap), and **Orders/Commands** (instructing an individual to manipulate physical entities) — these fall under “Thoughtful (Deliberate) Speech.” Separately, there is “**Rash**” (**Reflexive**) **Speech** — quick, unplanned reactions to upsetting, physical, or optical stimuli (e.g., yelling “Ouch!”).

#### 4. Can speaking move physical entities? And if yes, how?

**Yes.** If you speak a command to someone else (e.g., “Please move that box”), your speech act triggers their physical act — speech becomes a mechanism of **remote control over other actors**. This is the same principle behind AI commands and API calls: an AI generating a text string is performing an action that can move physical entities if that text is parsed and executed by a downstream robotic or software system.

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### Slide 17: Speech Act — Purposes

What purposes can a deliberate (“Thoughtful”) speech act serve?

Three: (1) **change information of another individual by a declaration** (storing/overwriting a fact in their mental database); (2) **get information from another individual by a question** (filling a gap in one's own database); (3) **order an individual to manipulate physical entities** — a physical-effect command, remotely controlling matter through another agent. This maps directly onto NLP **Intent Classification**: is the user declaring, questioning, or ordering?

What distinguishes “Rash” (Reflexive) speech from Thoughtful speech?

Rash speech bypasses the deliberate planning loop entirely — it is a quick, automatic reaction to an upsetting, physical, or optical stimulus (like shouting “Ouch!” when touching a hot stove, or a system automatically logging an “Error!”), rather than a deliberately crafted message aimed at a goal.

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### Slide 19: Workshop “Action Theory” (3 brainstorming prompts)

#### 1. Which questions should an action theory answer?

Per Slide 20's own answer: an action theory should explain **why individuals do what they do**, and — in the social domain — **how persons can live and work together**. More granularly, it must define what an action *is* (a change in the physical or mental world, always combined with motion), and specify **why and when** an action is triggered (per Slide 20: when perceived impressions do not correspond to inner values).

#### 2. How can action theory be connected to an information theory?

Per Slide 23's triangle of interconnections: **Information is what will be produced out of action** (an action changes the physical world, generating new sensory data — an Impression — which is new Information), and conversely, **Action is the source of information** — the two are locked in a continuous feedback loop rather than being independent theories.

#### 3. How can it be connected to a thought theory?

Also per Slide 23: **Action is what thinking drives; Action is the output of thinking** — thought is the engine that ingests Information and outputs Action — while **Thinking is massively influenced by action**, since executing an action changes the world, generating new impressions that thinking must then process. Action, Thought, and Information form a single, indivisible, continuous loop (an Agentic Feedback Loop), not three siloed theories.

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## Slide 20: Action Theory — Thesis

### What is this theory needed for?

To explain **why individuals do what they do**, and — in the social domain — to explain how persons can live and work together.

### What is an action?

A **change in the physical or mental world**. Actions always create, change, or delete entities, and action is always combined with motion/movement.

### Why and when is an action started?

**When perceived impressions do not correspond to inner values** — i.e., when the agent's current model of reality mismatches its internal goal state. This is functionally identical to a control-loop/PID-controller or a reinforcement-learning reward signal: perceive a state, compare to a target, and act only when there's a discrepancy to close.

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## Slide 21: Action Theory — Purpose of Action

### Is action the only connection between the physical world and the mental world?

**Yes** — explicitly stated as the module's core philosophical thesis: *“Action is the only connection between the physical world and the mental world.”* Since Body-Mind Dualism treats the mental world (thoughts, schemas, databases) and the physical world (atoms, geography, bodies) as entirely separate domains, Action functions as the “GLUE” between them — the physical world acts on the mind via Impression (e.g., light hitting an eye), and the mind acts on the physical world via Expression (e.g., a hand moving a box). A disembodied intelligence (an LLM with no APIs) is, by this logic, fundamentally cut off from reality until it is given the ability to execute physical or speech acts.

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## Slide 22: Action Theory — Fundamentals

**What is Action? → Manipulation of Entities.** Confirms the core definition: acting means manipulating (creating, changing, or deleting) entities in the physical or mental world.

**How is the Need for Action Determined? → When Perceived Impressions do not Meet Values.** Restates Slide 20's trigger condition: a mismatch between what is perceived and the agent's internal target values.

**Who Decides about the Necessity of Manipulation? → The Brain.** Establishes a strict hierarchy: actuators (hands, speech organs, motors) cannot decide to act on their own — the centralized brain/processor is the sole decision-maker.

**What is to be Done before Acting? → Producing an Action Plan.** For any thoughtful (non-reflexive) action, the brain must generate a plan before the physical or speech movement is executed — directly setting up Section 6’s “Action Plan” topic.

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## Slide 23: Action Theory — Part of Three Phenomena

### How are Action, Thought, and Information related to one another?

They form a single closed triangle/loop, not three separate theories: **Thinking generates information**, and **Information is what thinking works on**; **Action is what thinking drives and is the output of thinking**; **Action is what can be seen from thinking** (its only external expression) and, in turn, **Thinking is massively influenced by action**; **Information is what will be produced out of action** (via Impression) and **Action is the source of information**. This is the master architecture of an **Agentic Feedback Loop**: observe (Information) → decide (Thought) → act (Action) → observe the result (new Information) → repeat.

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## Slide 25: Workshop “Action Plan” (worked example: leaving a table tennis game for a lecture)

### 1. What values were not fulfilled?

The internal value “I must attend/pass this class” was not being met by the current perceived state “I am playing table tennis instead of attending the lecture” — a mismatch between goal-state and perceived reality (per Slide 20’s trigger condition).

### 2. From what was the action plan derived?

From the agent’s existing spatial and temporal knowledge schemas — knowing where the lecture hall is located, and knowing how long it takes to walk there — i.e., the action plan is derived by drawing on stored, previously-learned representations (Module C/D’s Representation and Space/Time frameworks) rather than being invented from nothing.

### 3. What was the content of the action plan?

A sequence of atomic steps combining speech and physical acts: put the paddle down, say goodbye to the opponent (a speech act), walk down the hall (a physical act), sit at the desk — a concrete, ordered trajectory from the initial state (playing table tennis) to the final/wanted state (seated in the lecture).

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## Slide 26: Action Plan — Thesis

### What is an Action Plan, and how is it executed?

An action plan is always produced in the mind by thinking; it shows how to transform a **discourse world** (see *Session C*) from an **initial state** to a **final state**. Every single atomic step of motion or speech is a node on this planned trajectory, and the plan’s actual “processing” — its execution — is carried out physically through motion and speech. This is architecturally identical to **state-space search** in AI (e.g., A\* search or an RL policy): the world is represented

as a state graph, the “Action Plan” is the optimal path found between states, and the actuators simply execute the steps along that path.

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## Slide 27: Action Plan — Planning and Execution

**What is the complete lifecycle of an action, from perception to result?**

Four stages, forming a continuous loop: **(1) Sense** — the **Current World** is observed and projected inward as the **Perceived World**; **(2) Evaluate** — the brain compares the **Perceived World** against its internal **Values**, and any mismatch generates the **Action Plan**; **(3) Plan/Act** — the **Action Plan** triggers a **Change**, initiating **Execution Start** targeting the **Wanted World**; **(4) Result** — as execution completes (**Execution End**), the real world is altered and becomes the **Adapted World**. Successfully implementing this full **Sense → Evaluate → Plan → Act** loop is, per the module, the definition of a functioning autonomous/Strong AI agent.

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## Slide 29: Conclusion

**What are the core takeaways of the “F Action” module?**

- (1) **Prerequisites**: proper action requires planning and anticipation — thinking into the future, not just reacting;
  - (2) **Taxonomy**: action covers exactly three areas — Physical act, Speech act, and Thinking act;
  - (3) **Motivation**: action theory explains why action is started — a mismatch between values and impressions;
  - (4) **Structure**: action plans are the foundational basis for implementing or executing any action;
  - (5) **Synthesis**: Action is the **core of intelligence** — it is what actually combines all aspects of the brain (perception, thought, knowledge) together into something useful, meaning neither a massive database (Information) nor a powerful logic engine (Thought) alone constitutes intelligence without it;
  - (6) **Interconnectivity**: Action is inseparably bound to Information and Thinking in the continuous feedback loop described in Slide 23 — true AGI requires the full closed loop of observe → think → act → observe-again, not any one piece in isolation.
-

## Session G — Concept

### Slide 2: Repetition — “Action” (Review of Module F)

#### What is an action?

Per Module F (see *Session F*): a **change (manipulation) of an entity** in the physical or mental world, always combined with motion/movement.

#### Which basic action types do you know?

**Physical acts** (moving things), **speech acts** (moving information), and **mental acts** (moving thoughts) — the three verbs Module F organizes into Outward (physical, speech) and Inward (mental) directions.

#### What is an action theory?

A framework explaining **why individuals do what they do**, and how persons can live and work together — formally tying action to a trigger condition (a mismatch between perceived impressions and inner values) and to the Work System model from Module A.

#### Why do individuals start an action?

Only when they detect a **mismatch between their perceived impressions and their inner values** — i.e., when the current state of the world (as perceived) diverges from the agent’s internal goal/target state.

#### What is an action plan and what components does it have?

A mental trajectory, built by thinking, that transforms a discourse world from an **initial state** to a **final (wanted) state**. Its components are the individual steps of motion and speech that carry out that transformation — the plan is the calculated path; motion and speech are its execution/processing.

This review sets up the exact problem Module G solves: **how does the brain actually build that action plan?** Answer: it builds it using **Concepts**.

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### Slide 3: Workshop “Concept” (8 brainstorming prompts)

#### What is not a concept?

A single, raw, unconnected data point with no schema behind it is not (yet) a concept — e.g., a bare physical object by itself, or an isolated name/number with no associated network of expectations, is just a **Primitive Concept** or **Entity** at most, not a full Concept. A concept requires a **schema** — a structured network of associations/expectations — layered over entities; the raw entities or atomic labels underneath that schema are not concepts in themselves.

#### What is a concept?

Per the module’s own eventual formal answer (Slide 20 onward): a Concept is a **“Schema Carrier”** — an abstract template/network of associations that defines what a class of entities is expected to be or do, independent of the specific abstract or physical thing it’s applied to. It’s the same underlying data structure whether applied to something physical (a building) or something entirely abstract (time) — proving a concept is a universal, domain-independent cognitive/computational structure.

### Find and describe an example of a motion concept.

E.g., “walking”: a schema expecting alternating leg movement, forward displacement over time, and ground contact — the concept doesn’t depend on *who* is walking; it’s the abstract template that any entity performing that motion instantiates.

### Find and describe an example of a building concept.

E.g., “house”: a schema expecting delimited walls, a roof, an entrance, and the “is a” role of providing shelter — the physical structure itself is not the concept (per Slide 9’s distinction); the concept is the human-defined bordered category applied to it.

### Find and describe an example of a time concept.

E.g., “morning”: a schema expecting a position of the sun/earth-rotation within a specific range, associated expectations (waking, breakfast) — reusing Module D’s definition of time as a mental concept applied to a specific recurring interval.

### Find and describe an example of a space concept.

E.g., “room”: a schema expecting a bounded volume with walls/floor/ceiling, an inside/outside distinction, and expected contents (furniture) — reusing Module D’s space-as-mental-concept framework, applied to a bordered volume.

### Find and describe an example of a thought concept.

E.g., “doubt”: a schema expecting a mismatch between two competing internal representations/values, with an expected behavioral consequence (hesitation, further inquiry) — a purely mental-world concept with no direct physical counterpart.

### Describe the concept “mother.”

Per the module’s own detailed example (Slides 14–15): “mother” is not a standalone property but a **Native Concept** — one half of a **counterpart role pair** (mother & daughter/son) generated by an association (an edge) between two entities. An entity “is” a mother only in relation to a specific child-entity it is linked to (Slide 15, assertion 9: “Cornelia can only be called a mother if she has a daughter”) — the concept is carried entirely by the relationship, not by any property intrinsic to the entity alone.

---

## Slide 6: Entity — Definition and Delimiters

(*This restates Module C, Slide 16 — see Session C\* for the original context; the module deliberately re-anchors the same foundational definitions before building the new Concept architecture on top of them.*)\*

**What Characterises an Entity in Principle? → It is delimitable.** The one universal requirement: if you cannot draw a boundary around something, it is not an entity.

**What are Delimiters? → e.g., surfaces, different perceptions.** Physical boundaries are set by material interfaces (surfaces) or by where a perceiving mind registers a discontinuity.

**What is a Physical Entity? → A conglomerate of matter and waves.** A bordered chunk of physical stuff.

**What is a Mental Entity? → A representation.** An internal, conceptual boundary rather than a physical one — bounded by its role as an internal mapping of something.

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## Slide 7: Entity — Nature

### What natures (ontological categories) can an entity have?

Three: **Physical** (exists in the physical world, e.g., the actual Eiffel Tower), **Mental** (exists in the mental world, e.g., the thought of the Eiffel Tower), and — newly introduced here — **Virtual** (exists only as a mental fabrication with *no physical counterpart at all*, e.g., a dragon, a fictional story, a lie). The mind manipulates virtual entities using the exact same cognitive machinery as real ones, which is the foundation of generative AI, simulation, hypothesis, planning, and art — an agent must be able to construct and reason about things that don't (yet, or ever) exist physically.

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## Slide 8 / 9: Entity — Primitive Concepts

### What are the primitive concepts a mind uses to represent an entity, and what are they, precisely?

Five primitive concepts, defined mechanically (Slide 9): **name** (a human-created phonetic/textual pattern used to identify entities and move information between individuals), **number** (the observed frequency/quantity of entities), **time** (the position of earth's rotation relative to the sun — the base for dynamic/temporal logic, per Module D), **role** (an action/behavior expectation, serving as a conceptualisation anchor), and **variable** (a placeholder needed for thinking and computing). A sixth item, the **physical entity** itself, is explicitly *not* a concept — it is the human-defined bordered physical area/volume that serves as the base for static and spatial logic (per Module D). The module's key claim: these primitive concepts are structurally identical to **primitive data types in programming** (strings, integers, booleans/roles, variables) — i.e., the human mind's basic representational building blocks and a compiler's basic data types are the same underlying architecture.

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## Slide 10: Entity — Role as an Action or Behavior Concept

### What is a “Role,” conceptually?

A Role is not a static noun or label — it is an **action (active) or behavior (passive) expectation** attached to an entity. Assigning the role “teacher” to an entity attaches the active expectation “this entity will teach” or the passive expectation “this entity will be listened to.” In OOP terms, a Role maps to the **methods/functions** attached to a Class — an object doesn't just hold static data, it holds behaviors expected to be performed or received.

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## Slide 11: Entity — Variable

### Where and when do we use variables in thinking or speaking?

Constantly, whenever an entity slot in a thought or sentence is required but not yet specifically filled — e.g., saying “Someone should fix that” uses “Someone” as a mental variable: the brain knows an entity is needed to complete the action but hasn't mapped a specific individual to that

slot yet, holding the placeholder open so the rest of the thought/plan can still be processed. This confirms human thinking relies on algebraic placeholders exactly the way a software compiler does — without the ability to hold variables, an agent couldn't construct plans or reason about inputs it doesn't yet know.

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### Slide 13: Workshop “Association” (4 brainstorming prompts)

**What happens very automatically in our brain, when we think about two entities or if we are confronted with two entities at the same time?**

The brain does not log them as two isolated, unrelated variables — it **immediately and automatically tries to connect them**, generating an edge (an association) between the two nodes, inferring a relationship (e.g., “the robot works inside the building”) without being explicitly told to.

**Can we link any arbitrary objects in a meaningful way?**

Yes, though the “meaningfulness” and specific role-pair generated depends entirely on context — any two entities presented together will trigger the brain's associative machinery to propose *some* relational hypothesis, even if it must be a loose or speculative one (proximity, function, causality, narrative), because isolated, disconnected data points are cognitively useless on their own.

**Why is your brain called to work associative?**

Because it is fundamentally a **network**, and networks derive meaning purely from connections (edges), not from isolated nodes — a brain that only stored atomic, disconnected facts without linking them could not generate context, inference, or understanding.

**What do you think, if you are confronted with the entity “Cairo Building” and the entity “WÜbota” (a humanoid robot)?**

The mind automatically proposes a connecting relationship even between two entities with no obvious prior link — e.g., “the robot works inside the building,” “the robot is being demonstrated at the building,” or “the robot was built to operate in structures like this one.” The exact edge generated is less important than the demonstration that association-generation is automatic and near-instantaneous whenever two entities are perceived together.

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### Slide 14: Association — Counterpart Paradigm

**What rules govern how the mind links two entities together?**

The **Counterpart Paradigm**, with four rules: (1) linking two entities always assigns each of them a **compatible Role**, forming a role pair; (2) roles are compatible when each entity's role describes what it *is* for the other entity, and vice versa; (3) each role represents that entity's action/behavior expectation; (4) the two roles of a single association must always be **counterparts** — their expected behaviors must fit together without contradiction or deadlock (e.g., Thrower/Catcher, Mother/Daughter) — an association where the two roles don't logically complement each other simply fails to form.

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## Slide 15: Association — Worked Example (Cornelia and Julia)

### What does the “mother & daughter” example prove about how roles work?

It proves that a role like “mother” **cannot exist in a vacuum** — it is generated strictly by the association (the edge) between two entities, not by any property intrinsic to one entity alone. The ten logical assertions establish: Cornelia is Julia’s mother and Julia is Cornelia’s daughter (inverse pair); together they form a mother-daughter relation with each holding the corresponding role; the two roles (mother/daughter) are counterparts and their pairing defines the *type* of the association; both roles carry compatible action expectations; and crucially, **Cornelia can only be called a mother if she has a daughter, and Julia can only be called a daughter if she has a mother** — the role is entirely relational, not a standalone attribute. This is exactly the Subject-Predicate-Object logic of Semantic Triples in Knowledge Graphs (RDF/SPARQL): `<Cornelia> <is_mother_of> <Julia>` automatically guarantees the inverse `<Julia> <is_daughter_of> <Cornelia>`.

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## Slides 17–19: Association Network

### How does an Association Network scale from a single relationship into a full “Family” or social structure — and what is the “is a” relationship?

By chaining many Counterpart Associations (Slide 14) between many entities, so that a single entity ends up holding multiple roles simultaneously, depending on which edge is being traversed — Entity A can be a Mother via one edge and a Daughter via another edge in the same network (Slide 17); no entity in the graph has an actual human name, only role-defined edges. On top of this, the special “**is a**” relationship (Slide 18) connects a specific entity to a broader external classification (e.g., Entity A “is a” Pilot, or “is a” Student) — this is the mechanism of **inheritance and polymorphism** in object-oriented terms: once an entity “is a” member of a class, it inherits all the behavioral expectations of that class in one step. Crucially (Slide 19), the network is infinitely expandable using the exact same universal logic — adding a University (Entity F, counterpart role “Student”) or a Company (Entity G, counterpart role “Pilot”) uses identical Counterpart-Association math to the biological family edges, proving that biological, educational, and economic relationships are all instances of the same underlying graph architecture. This is directly why, if the AI already knows **A is Wife of E** and **A is Mother of B**, it can algorithmically deduce **E is Father of B** purely by graph traversal.

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## Slide 21: Concept — Primitive, Native, and Derived

### What are the three tiers of conceptual complexity?

(1) **Primitive Concept** — a mental entity that stands for itself alone, not defined by any association schema (e.g., the raw name “Cornelia” — just a static string with no relational meaning on its own). (2) **Native Concept** — defined by a single association schema, usually mirroring a fundamental physical reality directly (e.g., the counterpart link “mother & daughter,” which reflects biological reproduction). (3) **Derived Concept** — a network scheme built from *multiple* chained association schemas (e.g., “brother & sister,” which is not a direct physical/native relationship but a mathematical consequence of several native parent-child links combined, per Slide 23). In software terms: Primitive  $\approx$  a primitive data type (String, Int); Native  $\approx$  a simple key-value pair/tuple; Derived  $\approx$  a complex Class/Object containing nested relationships.

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## Slide 22: Concept — Primitive Concept “Name”

**What is a Name?** → **A phonetic/textual pattern.** It has no inherent truth or physical reality of its own.

**When is it Used?** → **When Individuals Communicate.** Names exist specifically to support inter-agent communication.

**What is it Used for?** → **To Assign it to Mental Entities.** A name is attached to a mental entity as its label/pointer.

**Why is it Used?** → **To Shortcut the communication.** Rather than transmitting an entity’s entire complex association network (all of Cornelia’s relationships, roles, and history) every time she is referenced, the sender packages all of that graph data into a single string (“Cornelia”) — a name functions like a **pointer or hash**: a small token standing in for a much larger, complex mental structure stored elsewhere.

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## Slide 23: Concept — Derived Concept Algebra

**How is a derived concept like “brother & sister” mathematically constructed?**

It is proven to be an “illusion” in the sense that it is a semantic shortcut for a longer graph traversal, not a direct native relationship. Formally (using the module’s own **GREBTL** notation): {<sister> (sister & brother) <brother>} is algebraically **equal to** {<sister> (daughter & mother) <mother> (mother & son) <brother> (son & father) <father> (father & daughter) <sister>} — i.e., traversing from <sister> up to a shared parent (<mother> or <father>) and back down to <brother> mathematically proves the sibling relationship. The brain (or an AI) takes that longer graph loop and creates a new, direct shortcut edge (“brother & sister”) to save processing time on future lookups — exactly how graph-query inference (e.g., SPARQL or Cypher queries for a shared-parent pattern) automatically derives a “sibling” relationship without it ever being explicitly stored.

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## Slide 24: Workshop “Family” (worked example)

**Find the concept schemas for three of the red (non-native/derived) concepts in the information network.**

Applying Slide 23’s traversal logic: **Grandmother** = the native path Mother → Daughter, where that Daughter is also a Mother to a further Child (i.e., “mother of my mother”). **Uncle** = the native path Brother → (shared parent) → Child, i.e., the brother of one’s parent. **Mother-in-Law** = the native path Wife/Husband → (spouse’s) Mother, i.e., the mother of one’s spouse. In each case, the “red” derived relationship is not separately stored — it is calculated on demand by tracing a specific chain of native (blue) parent/child/spouse edges, which is exactly why efficient knowledge-graph engineering favors computing such derived roles via inference rather than redundantly storing every possible relationship label.

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## Slide 25: “Why is a horse not a horse?”

### Why is a (mental) “horse” not a (physical) horse?

Because the physical animal and the mental concept live in strictly separate domains (per Body-Mind Dualism). The physical horse is just a conglomerate of matter and waves; the brain does not — and cannot — store that animal itself. Instead, it stores a complex **Schema**: a network of expectations about size, shape, and behavior. When the mind needs to represent the specific physical animal, it creates an internal **Entity** placeholder and assigns it the **is a** role of “horse,” linking that entity to the schema of expectations — so “horse” (mental concept/role) and the horse (physical entity) are never the same thing; one merely represents and predicts the other. This is exactly why an AI camera doesn’t “see” a horse — it sees pixels, instantiates a bounding-box Entity, compares it against stored schemas, and assigns an **is a** classification with a confidence score.

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## Slide 26: Workshop “Horse” (2 prompts)

**Find compatible counterpart roles for the role horse (think about what a horse is linked to in real life).**

Per the Counterpart Paradigm (Slide 14): if the horse holds the role “Mount,” its counterpart is “Rider”; if it holds the role “Prey,” its counterpart is “Predator”; other plausible pairs include “Draft Animal” ↔ “Handler,” or “Racehorse” ↔ “Jockey” — each pairing must satisfy the non-contradictory, complementary-behavior requirement.

**Define a “main body part” concept for a horse, describing the main parts and their position relative to each other.**

An internal association network mirroring anatomy: Head connected to Neck, Neck connected to Torso, Torso connected to Legs (and Tail) — proving that even a single “atomic-seeming” entity like a horse is, on closer inspection, itself a nested concept made of smaller associated sub-entities (exactly the graph structure needed to rig a 3D physics model or generate a semantically consistent image/video of a horse).

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## Slide 28: Conceptualisation — Learning

**What is Conceptualisation, and how does Learning of new concepts happen?**

**Conceptualisation** means defining a concept together with its schema; **Learning** means internalising concepts into the mind. The module lists four modalities of learning: (1) **out of experience** (direct, unsupervised exposure to patterns in raw data), (2) **through repeating action patterns** (trial-and-error, i.e., reinforcement learning), (3) **through combination and/or adaptation of known concepts** (transfer — e.g., combining “horse” and “horn” schemas to construct “unicorn”), and (4) **by other individuals like teachers** (direct, supervised instruction, i.e., supervised learning / RLHF). These map directly onto the four major branches of machine learning: unsupervised, reinforcement, transfer, and supervised/human-feedback learning, respectively.

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## Slide 29: Workshop “Conceptualisation”

How would you implement a conceptualisation software to get concepts out of received information networks — e.g., a system that learns the concept “sister & brother” on its own?

The software would need to continuously scan the information/association network for **repetitive geometric patterns** rather than being told the answer directly: if it notices that any two nodes sharing the same two parent nodes consistently interact in a specific, recurring way, the algorithm should automatically draw a new shortcut edge between them and label it “sibling” — essentially automating the exact traversal-and-shortcut process described in Slide 23, but discovering the pattern itself rather than having it hand-coded. This is presented as the “holy grail” difference between current, largely-static LLMs (which know “sister” only because humans supplied labeled examples) and a true AGI capable of **unsupervised schema generation** — inventing genuinely new concepts by recognizing novel structural regularities in raw relational data.

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## Slide 31: Materialisation (Instantiation)

What is Materialisation, and how does an abstract schema become concrete reality?

**Materialisation (instantiation)** means bringing a concept into real life either by **Acting** (in the physical world) or **Thinking** (in the mental world). The mind holds abstract, empty templates (Concept Schemas, with unfilled variable slots like <sister>, <brother>, <father>, <mother>); when an Actor decides to act, it “instantiates” that template by plugging specific, real entities into those empty variables — e.g., the abstract family-structure schema becomes concrete when <sister> is replaced with **Julia**, <brother> with **Caesarion**, <father> with **Julius**, and <mother> with **Cornelia**. This is exactly **Object Instantiation** in software engineering: a Class (the abstract schema) cannot itself be interacted with — a constructor must instantiate a specific Object from it before it can exist in active memory and execute actions.

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## Slide 33: Conclusion

What are the module’s core takeaways?

**Base:** Concepts and their schemas are built on top of association and information networks — you cannot have a Concept without first having an Association Network beneath it. **Nesting:** concepts nest in different forms — a concept can be part of a larger overlaying concept (simple nesting), or interconnected with other concepts (complex nesting) — mathematically identical to classical inheritance in object-oriented programming. **Core:** Concepts are the core of both learning and acting — you cannot learn (acquire new schemas) or act (materialise schemas into reality) without them. **Key:** Concept schemas are the key not only to intelligence but also to **curiosity** — curiosity is simply the algorithmic drive to find the missing variables needed to complete a schema (directly linking back to Module E’s Knowledge Gap mechanism). The module’s ultimate claim: building AGI does not require inventing a new computational paradigm — engineers already have the necessary tools (object-oriented programming, graph databases, pointers, inheritance); what’s needed is arranging those existing tools into the continuous loop of perception → schema generation → instantiation described across this module.

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## Session H — Memory

### Slide 2: Repetition — “Concept” (Review of Module G)

#### What is a primitive, native, and derived concept?

Per Module G (see *Session G*): a **Primitive Concept** is a mental entity standing for itself alone, undefined by any association schema (e.g., a raw name). A **Native Concept** is defined by a single association schema, usually mirroring a physical reality directly (e.g., “mother & daughter”). A **Derived Concept** is a network built from *multiple* chained association schemas (e.g., “brother & sister,” mathematically derived from combined parent-child links).

#### What is the counterpart paradigm?

The rule (Module G, Slide 14) that linking two entities always assigns each of them a **compatible Role**, forming a role pair — the two roles must be counterparts whose expected behaviors fit together without contradiction or deadlock (e.g., Mother/Daughter, Thrower/Catcher).

#### What is not a concept?

A single, raw, unconnected data point with no schema behind it — an isolated entity or a bare name/number with no associated network of expectations is not yet a full Concept (see *Session G*, Slide 3).

#### How can concepts be presented?

As association-network graphs — nodes (entities) connected by role-pair edges (associations), scalable from simple binary links (native concepts) up to complex multi-edge networks (derived concepts), and instantiated into concrete reality via Materialisation.

#### Why is “horse” not a “horse”?

Because the mental concept (a schema of expectations about size, shape, and behavior) and the physical animal (a conglomerate of matter and waves) exist in entirely separate domains under Body-Mind Dualism — the brain never stores the animal itself, only a representational schema linked to it via an *is a* role (see *Session G*, Slide 25).

#### How are concepts helping us to be curious?

Concept schemas contain empty, unfilled variable slots; curiosity is simply the algorithmic drive to detect those gaps and seek out the missing information needed to complete the schema (Module G’s Knowledge Gap connection, and Module E’s original Knowledge Gap mechanism).

This review sets the stage for Module H’s question: now that we know *what* a concept is, we need to know **where and how it is stored**.

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### Slide 3: Workshop “Memory” (7 brainstorming prompts)

#### How and when do we remember experiences or impressions from the past?

Per the module’s later formal answer (Slide 13): we remember via three main triggers — a new impression matching a stored schema (**Impressions**), an active schema having an empty slot that needs filling (**Incomplete Concepts**), or one memory automatically pulling up adjacent, linked memories by traversing the association graph (**Associative recall**).

#### Which thoughts are coming from alone? And why?

Thoughts that arise without any new external impression — driven entirely by internal, endogenous processes such as detecting a knowledge gap in an active concept schema (Slide 15’s “Remembering by Uncomplete Concepts”) or by associative recall triggered by another internal thought rather than a fresh sensory input. They occur because the mind continuously scans its own internal models for gaps and connections, not only in response to the outside world (echoing Module F’s “endogenous loop,” Slide 14).

### How is the access to persisted information implemented?

Per Slides 6–11 and 31–33: as a read/write translation between the mental world and the physical world. Persisting (“writing”) means translating a Mental Entity (a thought) into a Physical Entity (a carrier — synapses, ink, magnetic storage); recalling (“reading”) means scanning that physical carrier and projecting the pattern back up into the mental world to reconstruct the thought. Concretely, the module implements this literally as a relational database (TABLE ENTITY, TABLE CONCEPT, TABLE NAME, TABLE ASSOCIATION), queried via associative/graph traversal.

### Is persisted information information?

Not while it’s merely sitting in physical storage — per Module E’s thesis (see *Session E*), information is strictly a **mental entity**; persisted data is a **Physical Entity (Carrier)** — magnetic states, ink, synaptic configuration. It only becomes information again once a mind reads/recalls it and reconstructs the corresponding mental entity (Slide 7).

### How can we remember things that we will do in the future?

By storing **Action Plans** as a dedicated memory segment (Slide 29–30) — the same persistence/recall mechanism used for past experiences applies to a plan’s future trajectory (an initial state → final state transformation, per Module F), which is simply persisted ahead of its execution and recalled when the moment to act arrives.

### In which parts or segments would you divide our memory?

Per the module’s own answer (Slides 27–30): **Short-term** vs. **Long-term** memory, and within long-term storage, dedicated buckets for **Episode Information** (experiences), **Reality Information** (physical-world facts), **Individual Reality Information** (subjective facts), **Value Information** (goals/ethics), **Action Plans**, **Motion Concepts**, and **Speech Concepts**.

### Which information would you put into which segment?

Following that same taxonomy: personal lived experiences → Episode Information; objective facts about the shared physical world → Reality Information; an individual’s own subjective beliefs/model → Individual Reality Information; core goals, ethics, and target values → Value Information (segmented and protected, since it drives all action-triggering per Module F); saved procedures for completing tasks → Action Plans; motor skill templates → Motion Concepts; and linguistic/communication templates → Speech Concepts.

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## Slides 6 / 7: Persisting and Recalling a Thought

### How is a thought physically persisted, and how is it recalled?

**Persisting (writing):** a thought is purely mental and ephemeral — it vanishes once you stop thinking about it. To save it, it must cross the boundary from the mental world down into the physical world, where it is stored as a **Physical Entity (Carrier)** — written on paper,

or encoded by the brain physically restructuring synapses and chemical states. **Recalling (reading)**: the reverse process — the mind cannot think using the physical carrier directly; it must scan that carrier (read a notebook, fire impulses through a specific synaptic pathway) and project the pattern back up into the mental world to reconstruct the original **Mental Entity (Thought)**. In computing terms: persisting is a `write()` to a hard drive/SSD (Physical World); recalling is a `read()` back into volatile RAM (Mental World) so the CPU can actually operate on the data.

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## Slide 8: Task — Primitive Concept Name and Name Association

### How does the mind persist a simple name-to-entity link?

It does not store the physical object itself — it stores a **Primitive Concept** (a text string, e.g., “Eiffel Tower”) and a separate associative edge linking that string to the mental entity of the object, categorized by a specific role (e.g., “english name,” alongside “french name,” “german name” for the same object). This is architecturally identical to a **key-value store or translation dictionary**: the core entity is stored once, with multiple string properties attached via labeled edges, queried by role when needed (e.g., retrieving the German name for a German-speaking user).

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## Slide 13: Reasons for Remembering

### What triggers the brain to actually execute a memory recall?

Three drivers: **(1) Impressions** — recalling persisted information because a new physical input matches a stored schema; **(2) Incomplete Concepts** — a detected missing variable in an active schema (a knowledge gap) forces an active search through past data to fill it; **(3) Associative recall** — one memory automatically pulls up other, linked memories by traversing the association graph. These map to a search engine’s three retrieval modes: pattern/similarity matching (vector search) for impressions, structured querying (SQL joins) for incomplete concepts, and graph traversal for associative recall.

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## Slide 14: Remembering by Impressions

### How does recognizing a physical object trigger memory recall?

The physical eye (or sensor) perceives the real object, generating a raw “impression” at the mental-eye boundary; the brain immediately runs a pattern-matching search against its persisted memory, and upon finding a schema whose geometry matches, it “remembers” the object, instantly loading its associated variables (names, facts) into active thought. This is exactly how computer vision with vector databases works: convert an image to an embedding (the impression), run a similarity search, and retrieve the associated metadata for the closest match.

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## Slide 15: Remembering by Incomplete Concepts

**How does a missing piece of information (without any new sensory input) trigger memory recall?**

Entirely internally: the mind is processing a schema (e.g., the Family Graph from Module G) and detects an unfilled slot — an entity’s identity is unknown even though its role relationships (daughter/mother/wife) are known. That empty variable creates a **Knowledge Gap**, and this cognitive discomfort alone (with no new physical impression at all) triggers an active memory search to find the entity satisfying all the intersecting counterpart roles — directly analogous to an SQL query with JOINS searching for a missing foreign key based on known relational constraints.

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## Slide 17: Workshop “Forgetting” (3 brainstorming prompts)

**What do we forget and what not?**

Per the module’s own algorithmic answer (Slide 18): we retain what is **frequently used (high F)** and **recently used (recent LT)**, and we forget what has **low frequency of use and a distant last-used time** — data with no ongoing utility is pruned, exactly like a cache eviction policy.

**How do we compensate for memory lapses?**

Through associative recall and reconstruction rather than perfect storage — the mind fills gaps by inferring from related, still-persisted associations (e.g., deriving a forgotten detail from surrounding context via the graph traversal/derivation logic of Module G), or by re-deriving lost information the same way a “derived concept” is recomputed from the native concepts that still remain (see *Session G*, Slide 23).

**How would you implement forgetting information in a digital memory?**

By tracking, for every stored node, a **Frequency of Use (F)** and **Last Time of Use (LT)** counter (Slide 18) — implementing exactly the **LRU (Least Recently Used)** and **LFU (Least Frequently Used)** cache-eviction algorithms already standard in computer science: flag and progressively degrade/delete the associative edges of nodes with low F and old LT to reclaim storage and processing capacity, treating forgetting as necessary garbage collection rather than a system failure.

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## Slide 18: Forgetting — Frequency and Last Time

**What is the literal algorithm the brain uses to decide what to forget?**

Track two variables per node: **Frequency of Use (F)** — how many times has this node been accessed — and **Last Time of Use (LT)** — how recently was it accessed. High F + recent LT → the brain protects the node; low F + distant LT (as shown for the unprotected “Entity E” in the module’s example) → the node is flagged for deletion, and its associative edges degrade and eventually vanish. This is precisely the mathematics behind LRU/LFU cache eviction, proving that biological memory optimization and digital cache optimization use the identical underlying logic.

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## Slide 20: Primitives — Name

(*This restates Module G, Slide 22 — see Session G\* — reframed here specifically in the context of memory indexing rather than communication.*)\*

**What is a Name? → Phonetic/Textual Pattern. When is it Used? → When Individuals Communicate. What is it Used for? → To Assign it to Mental Entities. Why is it Used? → To Shortcut the communication.**

In the memory context specifically: because a name is a primitive data type, it requires virtually zero storage and zero compute to retrieve, making it the ideal lightweight index/pointer for a massive, complex underlying schema.

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## Slides 21–25: Name — Tertiary Association and Its Collapse

**How does memory encode the *context* under which a name is valid (e.g., that “Eiffelturm” is specifically the German name)?**

Initially via a **Tertiary Association** (Slide 21): rather than a simple binary edge (Entity ↔ Name), the brain draws a third edge directly *into* the associative line itself, pointing to a context concept (e.g., “Language”), creating a 3-way dependency — the name is valid *only* while that context is active. Because a 3-way graph is computationally expensive to traverse, the brain then **optimizes/denormalizes** it (Slides 22–25): the context node (e.g., “German”) is collapsed directly into the edge’s own label, turning the expensive 3-way network back into a fast, flat, binary one — ending with simple, direct edges labeled **french name**, **german name**, **english name** running straight from the entity to each string. This mirrors **denormalization** in database design: a fully normalized schema (with a separate Language table requiring JOINS) is mathematically pure but slow, so engineers collapse the context directly into the column/edge label to optimize for read speed.

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## Slide 26: Workshop “Time and Space” (2 brainstorming prompts)

**How would you model time in an information network?**

By attaching **StartTime** and **EndTime** properties to every node or edge in the graph — directly implemented in the module’s own database schema (Slide 33’s **TABLE ASSOCIATION** includes **StartTime**, **EndTime** columns), letting the system track exactly when a given association or fact was/is valid.

**How would you model space in an information network?**

By attaching 3D coordinate properties (**X**, **Y**, **Z**) to entities, mapped relative to other entities — directly reusing Module D’s relative/absolute spatial-logic framework (see *Session D*) as literal database fields.

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## Slides 27–28: Segments — Value Memory

**How and why are an agent’s “inner values” (goals) stored in memory?**

Action Theory (Module F) assumes inner values detect the necessity of action — but for that to work, those values must themselves be **persisted** in memory, otherwise the agent could never refer back to them when evaluating a mismatch. Values are not hardcoded; they are formed and reshaped over time by impressions and experiences (guided experience = education). Crucially, because values dictate the behavior of the *entire* system, they cannot be mixed randomly with trivial facts — they must be **segmented**, stored in a distinct, clearly identifiable, heavily protected memory partition so the logic engine can instantly and reliably query them when deciding how to act. In AI agent design, this is the distinction between a **System Prompt/Reward Function** (segmented Value Memory, given overriding priority) and standard **Context Memory** (everything else) — you would never store a core safety directive in the same bucket as a pancake recipe.

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## Slides 29–30: Memory Segments Defined

**What are the functional segments of a mind’s memory architecture?**

A high-level split between **Short-term Memory** and **Long-term Memory**, with long-term storage further divided into specialized data buckets: **Episode Information** (personal experiences), **Reality Information** (objective facts about the physical world), **Individual Reality Information** (subjective facts/beliefs), **Value Information** (goals/ethics), **Action Plans** (saved procedures for tasks), **Motion Concepts** (how to move), and **Speech Concepts** (how to talk). This maps directly onto a **microservices architecture** with specialized databases: Short-term Memory  $\approx$  RAM/Redis cache; Episode Information  $\approx$  a time-series database (event logs); Reality Information  $\approx$  a graph/knowledge database; Value Information  $\approx$  configuration files/system prompts; Action Plans  $\approx$  stored procedures/executable scripts.

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## Slides 31–33: Implementation — Database

**Can this entire cognitive architecture actually be implemented with conventional software tools?**

Yes — the module’s climactic reveal is a literal SQL schema: `TABLE ENTITY (UUID, Name, Type)`, `TABLE CONCEPT (UUID, Name)`, `TABLE NAME (UUID, String Name)`, and `TABLE ASSOCIATION (UUID, Entity1, Concept1, Concept2, Entity2, StartTime, EndTime)`. The `TABLE ASSOCIATION` explicitly implements the Counterpart Paradigm (Module G) by mapping `Entity1` to `Entity2` through paired concept/role columns, while `StartTime/EndTime` fulfill the Time primitive from Slide 26. This proves, without ambiguity, that the entire “Association Network” and “Concept Schema” architecture developed across Modules G and H can be built with a standard Relational Database Management System — no exotic or biologically-inspired hardware is required.

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## Slides 34–36: Conclusion

**What are the module’s core takeaways?**

Memory is crucial for intelligence — it is the central place where all information is stored and retrieved, and nothing else in the seminar’s architecture (Concepts, Action Plans, Values)

can function without it. Associatively organized access to persisted information has major advantages over flat storage, since it enables both similarity-based (Impression) and gap-driven (Incomplete Concept) recall, plus fast graph traversal. The model of the persisting system must be architecturally **complete** — properly covering time logic and space logic (per Slide 26), not treated as optional metadata. Most importantly: **“Implementation can be done conventionally”** — building AGI-grade memory does not require quantum computers or biological neural substrates; standard SQL/graph database architectures are fully capable of hosting Strong AI’s memory layer, provided the schema (Entity, Concept, Name, Association, with time/space properties) is designed correctly.

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## Session I — Mind

### Slide 1: Repetition — “Memory” (Review of Module H)

#### What is the main task of a memory?

Per Module H (see *Session H*): to **persist** (write) and **recall** (read) data — translating a Mental Entity into a Physical Entity (Carrier) to save it, and translating it back to reconstruct the thought when needed.

#### Which basic reasons for remembering do you know?

Three: **Impressions** (a new input matches a stored schema), **Incomplete Concepts** (an active schema has an empty slot that must be filled — a Knowledge Gap), and **Associative recall** (one memory automatically pulls up adjacent, linked memories via graph traversal).

#### How would you implement forgetting?

By tracking, per node, a **Frequency of Use (F)** and **Last Time of Use (LT)** — pruning nodes with low F and old LT, functionally identical to LRU/LFU cache-eviction algorithms.

#### What is a name?

A human-created phonetic/textual pattern used to identify entities and move information between individuals — a lightweight pointer, not the entity itself.

#### What are names used for?

To assign a fast, low-cost label to a mental entity, so its full underlying schema doesn’t need to be re-transmitted or re-derived every time it’s referenced.

#### Why are names so important for remembering?

Because they act as a **shortcut**: since a name is a primitive data type, it requires virtually zero storage and zero compute to retrieve, making it the ideal lightweight index into a much larger, complex persisted schema.

#### How is the memory related to time logic and space logic?

Every persisted association can carry **StartTime/EndTime** properties (Time Logic) and 3D coordinate properties (Space Logic) — memory isn’t just *what* is known, but a structure that must also track *when* and *where* that knowledge was/is valid (directly implemented as literal database columns in Module H’s **TABLE ASSOCIATION**).

This review sets up Module I’s transition: if Memory is where data sits at rest, what is the processor that actively runs on that data? That processor is the **Mind**.

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### Slide 3: The Brain (Hardware Specs)

#### What does the biological hardware running the Mind actually look like?

86 billion neurons, 100 trillion connections, over 2 million years of evolution, a 35,000–100,000-year-old modern configuration, about 1.3 kg, running on just **20 Watts**. The 20-Watt figure is the module’s headline benchmark: the human brain runs the equivalent of a hundred-trillion-parameter network on less power than a dim lightbulb — proof that a hyper-efficient, biologically-scaled AGI architecture is physically possible, in stark contrast to the megawatt-scale server farms required to train and run today’s trillion-parameter LLMs.

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## Slide 4: Workshop “Mind” (8 brainstorming prompts)

### What is the mind?

Per the module’s own eventual answer (Slide 35–37): the central unit (the “Processing Core”) of the brain where thoughts exist and thinking takes place — the active CPU of the cognitive architecture, distinct from Memory (the storage/hard-drive layer, Module H).

### What is the main task of a mind?

To run the mind’s full processing pipeline: Understanding, Reasoning, Learning, and Initiating (Slide 5’s agenda) — loading static information from Memory into active, volatile “thought,” operating on it, and issuing commands back out into the world.

### What are tasks of the mind in respect to our daily life?

Continuously converting raw impressions and stored knowledge into consistent concepts (Understanding), deriving new insight from what’s already known (Reasoning), growing the knowledge base from experience (Learning), and triggering the concrete physical/speech/mental acts (Initiating) needed to get through ordinary tasks and decisions.

### What has the mind to do with writing?

Writing is the physical manifestation — the “logging” — of an active thought onto a persisted, physical carrier (Module H’s persisting mechanism); it is one of the concrete outputs the mind can Initiate (a Speech/Expression Act, per Module F).

### What is writing?

The act of translating a currently-active mental entity (a thought) into a durable physical medium (ink, a digital file) so it can be persisted and later recalled or communicated — the same write() operation described in Module H, Slides 6 and 8.

### What do we think about, when we are writing?

The specific thought(s) currently active in the “Mind’s eye” (Slide 6) that are being converted into words — writing forces sequential externalization of whatever entity/thought is currently highlighted in the mind’s active attention.

### What is a thought?

Per the module’s formal answer (Slide 6): **“Aware Information Just Processing”** — information that is not merely stored, but actively being run through the mind’s machinery right now.

### What is the difference between thought and information?

Information (Module E) is a static mental entity that can sit passively in memory; a Thought is that same entity *while it is actively being processed* — i.e., “loaded” into the mind’s active execution space (the “Mind’s eye”), the same distinction as data at rest (on disk) versus data actively being computed on (in a CPU’s registers/RAM).

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## Slides 6–7: Thought (Mental Substance)

**What is a Thought?** → **Aware Information Just Processing.** Information actively being run through the mind’s machinery, not merely stored.

**Where are Thoughts?** → **Mind’s eye.** A specific, localized locus of active attention, distinct from the broader, passive storage of Memory.

**What is the Smallest Possible Thought?** → **One Entity.** The atomic unit of thinking is a single delimitable entity (Module C/G) — you cannot think about “half” of a variable; you must load a complete, discrete entity to process it.

**How Many Thoughts Can We Have in Parallel?** → (left open)

An honest, unresolved question — a possible hint at a single-threaded bottleneck in human consciousness (only one “highlighted” entity in the Mind’s eye at a time), left open for the audience to debate. For AI, this maps directly to the size and single- vs. multi-threaded nature of a system’s active **context window**: while a computer can run many parallel processes, the core “aware,” coherent reasoning engine may still need to handle discrete entities largely serially.

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## Slides 8–9: Thought — Highlighted Information (the Castle Example)

**What does “thinking” actually look like mechanically, using the castle knowledge-graph example?**

The full castle graph (main door, side doors, towers, bridge, connected by front/back/left/right edges) represents everything persisted in Memory. A “Thought” is simply the active highlight — the pointer or cursor — landing on one specific node (e.g., the “main door,” then later shifting to the “bridge”). The graph itself doesn’t change during a thought; only the position of the active pointer does. This is a clean illustration of an **attention mechanism**: the full knowledge graph is like a model’s entire embedding/context space, and the highlighted node is like the specific entity currently receiving the most attention weight — thinking is the act of moving that highlight across the persisted network, exactly like a graph-traversal algorithm.

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## Slides 10–12: Understanding (Associating)

**What means understanding?** (*Workshop prompt*)

Per the module’s formal answer (Slide 12): **taking a description or explanation and using it to form a consistent concept** — successfully integrating new input into the existing Association Network (Module G) without contradiction.

**Give an example where you understood a topic the first time in your life. / Give an example of a not-understood topic.** (*Workshop prompts*)

These ask for personal introspection to surface the missing ingredient in a failed understanding attempt — typically, a “not understood” topic is one where the new description could *not* be consistently wired into any existing concept in the learner’s network (an unresolved Incomplete Concept, per Module H), whereas a first-time understanding is the moment a new description successfully closes that gap.

**If you understand a phenomenon, what can you use that for?**

Per the module's dependency chain: successful Understanding produces **Knowledge** (Slide 12), which per Module F/E is precisely what's needed to derive **purposeful acting** — so understanding a phenomenon is directly usable for prediction, planning, and action, not merely intellectual satisfaction.

### How is knowledge and understanding connected?

Knowledge is the **direct result** of Understanding (Slide 12): Understanding is the *process* (integrating a description into a consistent concept); Knowledge is the *product* — information that has been successfully wired into that consistent graph structure, as opposed to information sitting unintegrated and inert.

**What is Understanding?** → **Taking Description/Explanation to Form Consistent Concepts.** **What is the Result of Understanding?** → **Knowledge.** **What Information is Needed?** → **Information Related to the Concept that is not Understood.** **What is Changed by Understanding?** → **The Information Network of the Memory.**

Together, these four confirm that Understanding is not a separate mental event floating free of storage — it is a literal edit/expansion of the persisted Association Network from Module H, triggered specifically by information relevant to the exact concept that isn't yet understood (you can't understand in a vacuum; the missing input has to target the actual gap).

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### Slide 13: Understanding — Example: Time Machine

**What is a Time Machine?** → **Moving our Brain and/or Body in the Time.** A cleanly definable concept — the mind can build a consistent schema for it using the existing native concepts of Movement, Body, and Time (Modules D/F), logically connected with new edges.

**What Restrictions do Time Machines have? / How can you Build a Time Machine?**  
→ (left open)

Two deliberately blank (“...”) answers — an honest illustration of an **incomplete concept** (Module H, Slide 15): the mind fully understands *what* a time machine is, but has active, unresolved knowledge gaps about *how* to build one or what constrains it.

**Why are Time Machines not Built yet?** → **High Complexity! Lack of Technology!**

Not because the concept is incoherent, but because the concrete engineering path (implementation) remains unsolved — proving that **Understanding** (having a clear, well-formed concept) and complete **Knowledge** (having every actionable variable filled in) are not the same thing. This is also the origin of **creativity/generation** in AI terms: a system doesn't need to have physically observed something to “understand” it — it only needs to logically associate existing variables in a novel combination (exactly how generative AI produces images or descriptions of things that don't physically exist).

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### Slides 14–16: Reasoning (Decider)

**What is reasoning?** / **Give one example of reasoning.** / **What is the precondition to make reasoning possible?** (*Workshop prompts*)

Per the module’s formal answers below: reasoning is gaining new insight purely from what’s already known; an example would be inferring “A and C must be related” purely from already-known facts “A relates to B” and “B relates to C” (transitivity, no new observation needed); and its precondition is **Logic** — without a valid logical framework, “reasoning” would just produce arbitrary or contradictory claims.

**What is Reasoning? → Gaining new Insights/Concepts out of Known Concepts.** Reasoning operates entirely within the existing Association Network — deriving new insight without needing any new external impression.

**What is it Used for Externally? → Decision Making, Action Plans.** Reasoned conclusions feed directly into choosing between options and constructing Action Plans (Module F).

**What is it Used for Internally? → Learning.** New insights derived by reasoning become new material to be persisted — reasoning is what generates the content that Learning then writes to memory.

**Precondition? → Logic.** Reasoning cannot function correctly without a valid logical substrate underneath it (Section 5’s entire focus). In AI terms, this is the definition of an **inference engine**: given known graph relationships, independently derive new, previously-unobserved relationships (e.g., if A=B and B=C, generate the insight A=C) — but only reliably if the reasoning is constrained by valid logical rules.

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## Slides 17–18: Learning (Growth)

**What is Learning? → Enlarging the Knowledge Base (Known Concepts).** Mechanically, nothing more than writing the outputs of Understanding and Reasoning permanently into Memory — if Reasoning draws a new logical edge in active “RAM,” Learning is the `commit()` that permanently saves that edge to the storage database.

**Where Comes the Information From? → Self Experience or Others’ Experience.** Two sources: direct impressions (Module E’s exogenous “Perception” pathway) or communicated/taught information (Module E’s “Sharing” pathway).

**How is it Done? → Understanding and Reasoning.** Learning isn’t a separate mechanism — it is what happens when Understanding (consistent integration) and Reasoning (derived insight) are persisted rather than left in active, volatile thought.

**Where is the Learned Stored? → (Subjective) Memory.** Explicitly marked “subjective” — reinforcing Module C’s thesis that no two minds hold an identical representation, since what gets learned depends on each agent’s own unique experiences and internal reasoning. For AI, this is the precise distinction between **Inference** (Reasoning, which happens in a volatile context window and is lost when cleared) and **Training** (Learning, which requires permanently updating the model’s own weights/subjective memory database) — an LLM reasoning within a single conversation is not the same as that LLM actually learning from it.

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## Slides 19–23: Logic (Decider)

**What is logic? / What is it used for? / Where does it come from (or how do we learn logic)?** (*Workshop prompts*)

Logic is the rule engine that governs how the mind is permitted to draw edges during Understanding and Reasoning; it is used to guarantee that derived conclusions are valid rather than arbitrary or contradictory; and whether it is innate or learned is left as a genuinely open question by the module — mirroring the live AI debate over whether logical/compositional reasoning must be architecturally hard-coded (symbolic AI, rule engines) or can emerge purely from learned patterns (as claimed for large neural networks).

**What are the Basic Logics, and what are the Complex Logics built from them?**

**Basic Logics:** **Naming** (Module H’s pointer/shortcut mechanism), **Placeholder** (Module G’s variable concept, marking an information gap), **Abstraction** (Module E’s complexity-reduction tool, reused for inference), **Transitivity** (if A relates to B and B relates to C, then A relates to C — the exact mechanism behind Module G’s derived concepts like “brother & sister”), and **Exclusivity** (an entity cannot be in two mutually exclusive states/positions at once — reused from Module D’s space axiom). **Complex Logics** are compounds built from these: **Temporal logic** (Transitivity applied to time-ordering — before/after chains), **Spatial logic** (part-whole “is part of” relationships), and ultimately **Concepts** and **Action Plans** themselves.

**Who first formalized this kind of logic, and what did they establish?**

The Stoic philosopher **Chrysippus** (estimated 280–210 B.C.), credited with: differentiating object (entity) / meaning (concept) / name — a direct historical precedent for Module G/H’s entity-concept-name architecture; declaring statements true or false; distinguishing propositional, interrogative, request/command, and other non-propositional sentence types; using the fundamental logic units “and,” “not,” “if,” “else” (literally the vocabulary of Boolean logic and programming conditionals); creating the two-valued axiomatic propositional logic (five inference rules — modus ponens, modus tollens, and their disjunctive/conjunctive variants); and formalizing **Transitivity** (if A causes B and B causes C, then A causes C). This makes an explicit claim of continuity: the logical infrastructure required for a Strong AI’s reasoning engine was substantially specified by ancient philosophy roughly two millennia before modern computer science.

**What concrete vocabulary does each logic domain use?**

**Time Logic:** before, after, overlapping, intersecting (reusing Module D’s Relative Time Logic vocabulary). **Number Logic:** greater than, less than. **Space Logic:** is part of, left, right, front, back, up, down, overlapping, distance, is located of (reusing Module D’s Relative Space Logic 6-parameter model and Module G’s “is a”/“is part of” relations). **Value/Legal/Propositional Logic:** the Hypothetical Syllogism rules above. This is effectively a minimum predicate/operator library specification for a formal reasoning engine.

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**Slides 24–32: Applying Logic — The Castle, Missing View Point, Exclusivity, and Naming**

**How does the mind detect and resolve missing information in a knowledge graph, using the castle example?**

The castle graph (main door, side doors, towers, bridge) initially has an edge missing its front/back label between tower 1 and the bridge — an **incomplete concept** (Module H). Rather than re-observing it, the mind **derives** the missing relation using Transitivity, from the already-known surrounding directional chain — exactly like link prediction in a knowledge graph.

### “WHERE is the View Point?”

This question exposes a *deeper* missing variable: none of the relative directional labels (front, back, left, right) are meaningful without a defined observer position (reusing Module D’s requirement for a Relative Space Logic **Reference Entity**). The mind resolves this by introducing an entirely new node — the “**view point**” — and re-anchoring every existing relation to it. This is a critical lesson for any spatial-reasoning or robotic AI system: relative directional language is meaningless without an explicit, represented observer/anchor entity.

### What does Exclusivity teach us, and how far does it apply?

Per the module’s core statements: “I never saw one entity at different places at the same time” and “I cannot place myself where another entity already occupies space” — directly mirroring Module D’s core space axiom (Slide 18). The follow-up questions (*Where do we know it’s the same time? the same entity? What if an entity could be in two places at once? Does exclusivity apply beyond spatial issues?*) push the audience to notice that confirming “this is the same entity/time” is itself an inference (relying on Naming and identity-tracking), not a raw perceptual given — directly the **entity re-identification problem** any computer-vision tracking or SLAM system must solve. The final question is answered on the very next slide: yes, Exclusivity is fully general — a pure math function like  $y = f(x) = x \times x$  contains Exclusivity (each input deterministically maps to one output) but **no** temporal or spatial logic at all, proving Exclusivity, unlike Time/Space logic, is required everywhere, including pure computation.

### What does the Gulf of Mexico/Gulf of America example teach about Naming and Time Logic?

Naming requires **Exclusivity** (a name should point to exactly one entity in a given context) and **Temporality** (the name-entity association has an explicit valid time range — “Gulf of Mexico” valid up to 2025-01-19, “Gulf of America” valid from 2025-01-20 — directly implementing Module H’s Tertiary Association/**StartTime-EndTime** mechanism). Whether Naming also requires Spatiality is left open. This is a live, real-world demonstration that a name is not a permanent, timeless label but a time-bound edge in a knowledge graph — exactly the versioning problem any AGI maintaining a real-world knowledge base must solve whenever an entity’s canonical name changes.

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### Slides 33–34: Initiating (Motion Start-up Setter)

**What is Initiated?** → **Motion, Speech and Thinking Acts.** Exactly Module F’s three action types.

**Why is Something Initiated?** → **Impressions are Hurting our Values.** Restates Module F’s trigger condition (a mismatch between perceived impressions and inner values) in more visceral language — the discomfort of that mismatch is what forces action.

**What Helps?** → **Action Plans.** Having a pre-built plan (Module F, Slide 26) ready to execute, rather than improvising from scratch.

**Challenge? → Immediate and Intermediate Feedback by Senses.** The mind must continuously monitor its own unfolding action via sensory feedback to know whether the plan is succeeding or needs mid-execution correction — precisely the architecture of a **closed-loop control system** (e.g., a robot detecting low battery, reasoning a path to the charger, initiating movement, and continuously adjusting via camera/LiDAR feedback until the goal state is reached).

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## Slides 35–37: Conclusion

### What are the module’s core takeaways?

The Mind is the central unit (the “Processing Core”) of the brain where thoughts exist and thinking takes place. Its main tasks are Understanding, Reasoning, Learning, and Initiating. **Logic is a non-negotiable dependency** — without it, none of the other tasks could function correctly. The mind’s executive authority is to **issue commands to act**, closing the loop with Module F’s Action Theory. And critically, its job doesn’t end at initiation: the mind **permanently observes its own initiated commands** for control and correction, running a continuous feedback loop rather than a fire-and-forget execution model. Altogether, this gives computer scientists a full architectural spec for an AGI’s central executive: form and integrate concepts (Understanding) → derive new insight (Reasoning) → grow the knowledge base (Learning) → validate everything against a small set of foundational logical operators (Logic) → issue commands into the world (Initiating) → never stop watching the results (continuous correction).

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## Session J — Language

### Slide 2: Repetition — “Mind” (Review of Module I)

#### Why is our mind a time machine?

Per Module I (see *Session I*) and its use of Time Logic (Module D): the mind can freely traverse temporal states in its internal model — recalling “yesterday” from persisted memory, or projecting “tomorrow” via planning — moving its own attention/“Mind’s eye” across time-indexed mental entities without physically moving through time itself. It is a “time machine” in the sense that thinking can jump backward (recall) and forward (planning) across a represented timeline, even though the body cannot.

#### What is a thought?

“**Aware Information Just Processing**” — information actively being run through the mind’s machinery right now, as opposed to information merely sitting passively in memory.

#### What is the smallest possible thought?

**One Entity** — the atomic unit of thinking; you cannot think about “half” of a variable, you must load a complete, discrete entity to process it.

#### Why does reasoning need a logic?

Because without a valid logical framework, “reasoning” (deriving new insight purely from known concepts) would produce arbitrary or contradictory conclusions rather than valid ones — Logic is explicitly named as Reasoning’s precondition in Module I.

#### What are the main tasks of a mind?

**Understanding, Reasoning, Learning, and Initiating** — the four functional sections of Module I’s agenda, all underpinned by Logic.

#### Try to describe what curiosity is.

Per Modules E, G, and H’s Knowledge Gap mechanism: curiosity is the algorithmic drive that fires when the mind detects an empty, unfilled slot in an active concept schema (an “incomplete concept”) — the discomfort of that gap motivates the mind to actively search for or request the missing information needed to complete the schema.

This review grounds the transition into Module J: Language does not exist in a vacuum — it is the **output mechanism for the Mind**. Before a system can generate language (behavior), it must have a functional Mind executing logical reasoning underneath it.

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### Slide 3: Workshop “Language” (6 brainstorming prompts)

#### What is a language?

Per the module’s later formal answer (Slide 13): a system for the **materialization and mentalisation of information according to generally established and accepted rules (grammars)** — i.e., a rule-governed protocol for converting internal mental entities into externally transmittable signals, and back again.

#### What is a language used for?

Per Slide 13's "WHY?": in the narrow sense, **transporting information from one brain to another**; in the broader sense, **controlling the action and behavior of social-technical systems** — directly matching the module's subtitle, "Behaviour Maker."

### What different kinds of languages do you know?

Per Slide 6's taxonomy: **Natural languages** (human languages, animal languages) and **Artificial languages** (formal languages — programming languages, modelling languages, semantic networks — and constructed human languages like Esperanto), plus an open question about whether a purely internal "**thought language**" (mental entities and their associations, with no external expression at all) also exists.

### What criteria can be used to differentiate languages?

Along the axes the module develops across Slides 6–7: natural vs. artificial origin; the **medium** used (sound waves, photons/light, persistent physical materials); the **expression channel** (oral, body language, facial expression, sign language, pantomime, writing/painting); the **impression/perception channel** (auditory, visual, tactile); and the **written-form encoding system** (phonetic/alphabetic, syllabic, consonantal, logogrammic/word-and-syllabic, or picture-based scripts).

### What is the purpose of language?

Per Slide 10: three interlocking functions — **Communication** (externalizing internal thoughts to another mind), **Support for structuring thoughts** (e.g., abstraction — the act of formulating language forces the brain to clarify its own internal models), and **Influencing the way we think** (a feedback loop: linguistic feedback received from others alters the brain's own future internal logic/thought structuring).

### What is syntax, grammar and semantics related to languages?

Per the module's three dedicated sections: **Syntax** is the structural ordering of elements (Section 3); **Grammar** is the rule-set governing valid modifications and combinations of those elements, e.g., morphology (Section 4); **Semantics** is the mapping of those structurally-valid, rule-conformant symbols back onto actual meaning — grounded mental entities (Section 5). All three are modeled as successive compiler layers a linguistic signal must pass through, whether the "compiler" is a human brain or a programming-language interpreter.

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## Slide 6: Types of Languages

### What types of languages exist, and is there a "thought language"?

**Natural Languages** (human and animal languages, which evolved organically) versus **Artificial Languages** (formal languages like programming/modelling languages and semantic networks, plus deliberately constructed human languages like Esperanto) — placing something like C++ and English on the exact same theoretical spectrum. The module also raises an open, unresolved question: is there a purely internal "**Thought language**" — the raw associational network of mental entities inside the mind, prior to and independent of any external expression? This is effectively the classical "Language of Thought" hypothesis, and the module visualizes the Brain as containing the Mind and Memory, with the Mind further subdivided into sensory "screens" (Vision, Hearing, Balance, Touch, Smell), a Motion Center, and an explicitly highlighted **Speech Center** bridging internal thought and external expression.

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## Slide 7 / 8: Types of Expression and Impression

**What are the different channels and media through which language can operate, beyond spoken sound?**

Language is decoupled entirely from “sound” once broken into its I/O layers: **Expression** (motion-based outputs — oral speech, body language, facial expressions, sign language, pantomime, writing/painting/signs), **Impression/Perception** (sensory inputs — auditory, visual, tactile, e.g. Braille), and **written-form encodings** (phonetic/alphabetic scripts like Latin, syllabic scripts like Japanese, consonantal scripts like Arabic/Hebrew, logogrammic/word-and-syllable scripts like Chinese or road-sign symbols, and picture stories) — all riding on a **Medium** (sound waves through air, photons for visual/optical transmission, or persistent physical materials). The worked example is a red **STOP sign**: an artificial, photon-based, visual-medium language using a word/symbol script — proving that an autonomous robot reading a stop sign via camera (photon medium, visual impression, symbolic script) engages in exactly the same linguistic process as a human hearing a spoken “stop” command. Language, in this framework, is entirely **medium-agnostic**.

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## Slide 10: The Feedback Loop of Purpose

**How does language function as a feedback loop rather than a one-way output?**

Three linked functions: (1) **Communication** — externalizing internal thoughts so another mind can receive them; (2) **Support for structuring thoughts** — e.g., abstraction; the very act of forcing an internal, fuzzy thought into a rigid external language structure makes the brain clarify and organize its own internal model; (3) **Influencing the way we think** — receiving linguistic feedback from others loops back into and reshapes the brain’s own future internal logic. Visually: thought-structuring happens internally → gets pushed out as communication → loops back in as feedback → which reshapes future thought-structuring. This is the exact mechanism behind **Chain-of-Thought prompting** in LLMs: forcing a model to externally “communicate” its intermediate reasoning steps measurably structures and improves its internal reasoning accuracy, exactly as the feedback loop predicts.

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## Slides 11–12: Natural vs. Artificial/Formal Language Communication

**How does natural language communication work mechanically, and how is programming a form of language?**

**Natural language** (Slide 11) is modeled as a transport protocol: Brain A takes an internal mental concept, **materializes** it into a physical medium (sound waves or text — reusing Module E’s “communication is not information” thesis), transports it across that medium, and Brain B **decompiles** the physical signal back into its own internal mental concept. **Artificial/formal language** (Slide 12) is shown to be structurally identical: a human brain takes an internal mental “Action Plan” (Module F), translates it into a formal artificial language (e.g., Java code), and transmits it directly into a silicon CPU, which executes it against its own Memory. This is presented as a genuine conceptual leap: computer programming is not a metaphorical “language” — it is literally the same linguistic communication process as human speech, just

materialized into a different formal syntax and transmitted to a different kind of “brain” (a CPU), used specifically to achieve **Action Control** over that CPU’s memory and execution state.

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### Slide 13: Language as a Tool

#### WHAT (in a technical sense)?

Language is the materialization and mentalisation of information according to generally established and accepted rules (grammars).

#### WHY?

In the narrow sense, to transport information from one brain to another; in the broader sense, to control the action and behavior of social-technical systems.

#### WHO?

Used by intelligent systems — humans and animals — and increasingly by machines that are starting to behave like systems capable of using language themselves (e.g., NLP systems).

#### HOW?

In a rule- and convention-based way, following syntax, grammar, and semantics, according to established communication rules.

#### WHAT (in a Work System)?

Just a tool — but the single most important one in group work systems (Module A’s Work System framework), since it is what alters and synchronizes the information and knowledge held by different individuals, and is even the mechanism of educating minds.

This gives the definitive systems-engineering specification for language: an I/O tool used to synchronize state (knowledge) between distributed nodes (individuals) in a network (a group work system) in order to achieve coordinated action — which is also exactly the role of **NLP** in AGI: NLP is not intelligence itself, but the networking/synchronization tool an AI uses to keep its internal database aligned with human nodes.

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### Slide 14: Workshop “Semantics” (3 brainstorming prompts)

#### What is syntax, grammar and semantics, and what are the differences or how are they connected?

As established across the module’s three dedicated sections: **Syntax** is the structural ordering check (does the sentence/statement match an expected pattern, like Subject-Predicate-Object, or **if-then-else?**); **Grammar** is the rule-set that further constrains valid combinations within that structure (e.g., morphological agreement — plural subjects need plural verbs); and **Semantics** is the final mapping of the now-structurally-valid, rule-conformant symbols onto actual grounded meaning (mental entities). They are connected as successive, dependent compiler stages: a string must pass Syntax, then Grammar, before Semantics can even be meaningfully evaluated.

**Semantics is seen as the meaning of a word, sentence, or text. If so, what does the sentence “meaning of a word” mean?**

Per the module’s later formal answer (Slide 27): the “meaning” of a word is **what we think, or what is intended to be thought, when we hear or read that word** — i.e., the specific mental entity in the brain (Module I’s “Mind’s eye”) that becomes highlighted/activated upon encountering that word. So “the meaning of a word” refers to the process (and result) of a lookup: the word is an external key, and its meaning is the internal, associated mental entity it retrieves.

**What meaning has the sentence “The tomorrow of yesterday is the yesterday of tomorrow”?**

This is the module’s central worked puzzle, resolved fully in Slides 22–23: despite being syntactically and grammatically valid, the sentence reduces mathematically to  **$T = T$  (“Today is Today”)** — a pure tautology carrying **zero new information**. See the detailed derivation below.

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## Slides 16–17: Syntax in Natural and Formal Language

**What is syntax, and how does it work identically across natural and formal languages?**

Syntax is the structural blueprint check performed before any meaning is evaluated. In natural language (Slide 16), the example sentence “The tomorrow of yesterday is the yesterday of tomorrow” is checked against the expected **Subject–Predicate–Object** order (Subject: “The tomorrow of yesterday”; Predicate: “is”; Object: “the yesterday of tomorrow”). In formal/programming language (Slide 17), the exact same structural-checking process governs a statement like `if (day == TODAY) then day++ else day--;`, verifying the expected **If-Condition-Then-Else** block ordering. Syntax errors occur identically in both domains whenever this structural blueprint is violated — whether the “language” is spoken English or C++.

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## Slides 19–20: Grammar — Morphology and EBNF

**What is Grammar (morphology), and how does it validate a sentence beyond mere structure?**

Grammar defines rule-based constraints applied on top of valid syntax — e.g., **Morphology**, the rules for modifying words to express tense, number, or case (Slide 19). The worked example — “The tomorrows of yesterdays is the yesterdays of tomorrows” — is syntactically fine (still Subject-Predicate-Object) but grammatically broken: the subject is plural (“tomorrows”) while the predicate (“is”) remains singular, violating subject-verb agreement. In compiler terms, this is exactly analogous to a **type error**: the structure (syntax) is correct, but an internal rule (grammar/type constraint) is violated.

**How is grammar formally specified for both natural and artificial languages?**

Via the **Extended Backus-Naur Form (EBNF)** (Slide 20) — the standard metasyntax used to mathematically define a formal language’s grammar. The module shows that an English sentence pattern (`simple declarative sentence ::= subject predicate`

object) can be defined using the *exact same* EBNF notation as a computer conditional statement (conditional statement ::= "if (" condition ") then" statement ["else" statement] ";") — direct proof that natural language grammar and programming-language grammar are, formally, the same kind of algorithmic rule engine.

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## Slides 22–23: Semantic Mapping and Reduction (Does the Sentence Have Any Meaning At All?)

**How does semantic mapping work, and does “The tomorrow of yesterday is the yesterday of tomorrow” actually mean anything?**

**Semantic Mapping** (Slide 22) is the process of translating physical text tokens back into abstract mental entities: “tomorrow” maps to **day** + **amplification** (forward offset), “of” maps to **belonging**, “yesterday” maps to **day** (with a backward offset implied), and “is” maps to **identity** (equality) — the word “is” itself carries no physical weight; it’s purely a pointer to the logical concept of identity.

**Semantic Reduction** (Slide 23) then runs the actual math, anchored to “Today” (T): - Yesterday (B) = the day before today:  $B = T - 1$  - Tomorrow of Yesterday (A) =  $B + 1 = (T - 1) + 1 = T$  - Tomorrow (D) = the day after today:  $D = T + 1$  - Yesterday of Tomorrow (C) =  $D - 1 = (T + 1) - 1 = T$

**Result: the sentence reduces to  $T = T$ , i.e., “Today is Today” — a tautology.** The sentence is perfectly valid in both Syntax and Grammar, but it is **semantically empty**: it provides zero new information, despite sounding profound. The module explicitly flags this as the ultimate limitation of current LLMs: they are extremely capable at Syntax and Grammar (generating perfectly structured, rule-conformant text), but because they lack a grounded semantic “reality engine” to actually run this kind of underlying math, they can generate confident, grammatically flawless sentences that are logically or mathematically meaningless — i.e., **hallucinations**.

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## Slide 24: Workshop “Brain” (5 brainstorming prompts)

**What comes to your mind when you think of “day”?**

Per Slide 26’s formal answer: **Day** is defined as “all the happenings and movements of entities between the two moments when the Earth rotates around itself one time” — a mental entity anchored directly to the physical rotation of the planet (reusing Module D’s Time framework), not merely an abstract dictionary label.

**What is the meaning of “day”?**

The mental entity representing one full Earth rotation — the physical anchor that grounds every other relative time-word (today, yesterday, tomorrow) that gets defined in relation to it.

**What is the meaning of “yesterday” and “tomorrow”?**

Both are defined *relative to* “Today” (the day at which we are currently receiving impressions): **Yesterday** is the day before Today; **Tomorrow** is the day after Today. Formally (Slide 26): “yesterday (before & after) today; today (before & after) tomorrow” — i.e., these words only

have meaning as relational offsets around the observer’s current temporal position, exactly like Module D’s Relative Time Logic (with “now” as the floating reference point).

**What happens in our brain when we hear the sentence “The tomorrow of yesterday is the yesterday of tomorrow”?**

The brain runs the full Syntax → Grammar → Semantics pipeline: it verifies structural ordering (Subject-Predicate-Object), checks grammatical agreement, then performs the semantic-mapping-and-reduction math shown in Slides 22–23 — ultimately computing that both sides of the equation resolve to the same anchor point, Today (T), and recognizing the statement as a tautology.

**What is the purpose of that sentence, if it has no meaning?**

The sentence has no **semantic** value (it conveys zero new information — a pure tautology) but can still carry other, non-semantic purposes: it functions as a rhetorical or pedagogical device (exactly as used in this very module, to demonstrate the difference between syntactic/grammatical validity and true semantic content), or more broadly could serve poetic, playful, or attention-grabbing (pragmatic) purposes — illustrating that a sentence’s grammatical well-formedness and its actual informational content are two entirely separate properties.

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## Slide 26: Meaning and Time Logic

**How does the brain anchor relative time-words (day, before, after, yesterday, tomorrow) to actual meaning?**

By grounding them all in physical reality and a shared Time Logic framework (reusing Module D): **Day** is anchored to Earth’s physical rotation; **Today** is the day during which current impressions are being received; **Before** describes observations/impressions already present in memory relative to new incoming impressions; **After** describes newly-processing impressions relative to what’s already stored; **Yesterday** is the day before Today; **Tomorrow** is the day after Today. This yields the compact rule: **“yesterday (before & after) today; today (before & after) tomorrow.”** For Strong AI, the module argues this proves an AGI cannot just statistically predict the next token in a string — it must possess an actual **world model** in which “day” is understood to correspond to the physical rotation of a 3D sphere, and “yesterday” refers to specific, immutable data already committed to memory (Module H), not merely a token with high co-occurrence probability.

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## Slide 27: The Mechanics of Meaning

**What is the meaning of a word, mechanically, and what happens when the meaning is unknown?**

The meaning of a word is **what we think, or what is intended to be thought, when we hear or read that word** — specifically, whatever gets highlighted in the “Mind’s eye” (Module I). Mechanically, a word functions as an external lookup key; the brain effectively runs a query against its own concept store (`SELECT concept WHERE key = 'word'`, in the module’s own database metaphor). If a match is found, that mental entity is loaded into active attention. **If the lookup fails** — the word’s meaning is genuinely unknown — the brain simply retains the raw text string in temporary memory and, if **curiosity is strong enough** (Module I/H’s

Knowledge Gap mechanism), triggers a background learning process to resolve that gap, rather than immediately discarding the unrecognized token.

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## Slide 28: Conclusion

What are the module's core takeaways?

- (1) **Language is the tool for exchanging information and for controlling systems** — directly matching the module's subtitle, "Behaviour Maker."
  - (2) **Perceiving a sentence or word immediately starts a mapping process in the brain**, searching for the right mental entity for a word, or the right mental information network for a full sentence.
  - (3) **Semantics is precisely that mapping process** — and because performing this mapping requires an active, concept-processing engine, there is necessarily a **Mind (thinking)** operating behind semantics; language comprehension is not a passive lookup table but an active cognitive act, tying Module J's entire framework back to Module I's definition of the Mind as the "Processing Core."
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## Session K — Motion

### Slide 1: Title and Administration

**What does the subtitle “Causer of Time and Space” mean, and how does it relate to Module D?**

It introduces a deliberate causal inversion of Module D’s own subtitle (“Guarantors of Entities,” see *Session D*). Module D argued Time and Space are the mental frameworks needed to guarantee an entity can persist and stay distinct. This module goes one step further, in the opposite direction: Time and Space do not exist as independent realities at all — they are mental constructs the Brain invents specifically to measure and describe **Motion**. You only know “time” has passed because the sun *moved* across the sky; you only know “space” exists because you had to *move* your body to reach a wall. Motion, in this framing, is the primary, fundamental phenomenon; Time and Space are the descriptive tools built to track it.

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### Slide 2: Repetition — “Language” (Review of Module J)

**Which types of languages do you know?**

Per Module J (see *Session J*): **Natural languages** (human, animal) and **Artificial languages** (formal/programming languages, modelling languages, semantic networks, and constructed human languages like Esperanto), plus the open question of a purely internal “thought language.”

**What purposes of a language did we discuss?**

Communication (externalizing thoughts), support for structuring thought (e.g. abstraction), and influencing the way we think (a feedback loop where received language reshapes future internal thought-structuring).

**In which system is language a tool?**

A **Work System** (Module A) — language is the single most important tool for synchronizing information and knowledge between individuals (or machines) in a group work system, and even for educating minds.

**What is semantics?**

The process of mapping structurally-valid (syntax + grammar checked) symbols onto their actual grounded meaning — the specific mental entity that gets activated/highlighted in the mind upon encountering a word or sentence.

**Why is semantics a process?**

Because performing that mapping requires an active, concept-processing engine (Module I’s Mind) to execute a lookup — meaning isn’t a static property sitting in the word itself; it is the *outcome* of a cognitive act triggered by hearing or reading it.

**What is pragmatics?**

The purpose or use-value a sentence carries *beyond* its literal semantic content — what the sentence is actually being used *for* in context, independent of whether it conveys new factual information.

**Which sentence has no semantics but pragmatics?**

Module J’s central worked example: “**The tomorrow of yesterday is the yesterday of tomorrow**” — it reduces mathematically to the tautology  $T = T$  (semantically empty, zero new information), yet still carries pragmatic value as a rhetorical or pedagogical device (exactly how it was used within Module J itself).

This review sets up the transition into Motion: once a Mind uses Language to compile an Action Plan (Module F/I), it must actually **execute** that plan in the physical world — and executing a plan requires Motion.

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### Slide 3: Workshop “Motion” (6 brainstorming prompts)

#### What is motion?

Per the module’s later formal answer (Slide 17): **the change of location and/or orientation of things in the physical world** — decomposing into exactly the two primitives the module dedicates full sections to: Linear Motion (location change) and Rotational Motion (orientation change).

#### Which characteristics or properties would you assign to motion or movement?

Per Slide 5’s formal taxonomy: Distance, Velocity, Precision, Complexity, Interaction, and Intelligence (plus Mass, added as a seventh dimension on Slide 7).

#### Which of your assigned properties are the most relevant?

This depends entirely on the application — per the Motion Taxonomy radar chart (Slide 6), an industrial robot’s most relevant axes are Precision and Velocity, while a humanoid robot’s most relevant axes are Intelligence, Interaction, and Complexity — there is no single universally “most relevant” dimension; relevance is profile-dependent.

#### What basic types or categories of movement or motion would you distinguish?

**Linear (translational)** and **Rotational** motion — the two fundamental, mathematically independent motion primitives formalized in Sections 2 and 3.

#### Is there a mental motion beside the physical motion? Explain.

Yes — directly reusing Module C’s radical materialist axiom that there are no activities other than movement (see *Session C*, Slide 10): thinking itself can be modeled as motion, specifically the movement of the “highlighted” active entity across the mind’s association network (Module I’s “Mind’s eye”/castle-graph example, see *Session I*). Just as physical motion is a change of location/orientation of matter, mental motion is a change in which entity currently occupies the mind’s active attention.

#### Which machines are built for motion? Try to find basic types.

Per Slide 6’s taxonomy: **industrial robots**, **mobile robots**, **service robots**, and **humanoid robots** — four canonical categories occupying very different profiles across the six motion dimensions.

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### Slide 4: Agenda — K Motion

- 1. Taxonomy → *How is it*

- 2. **Linear Motion** → *Location Change*
- 3. **Rotational Motion** → *Orientation Change*
- 4. **Part-Whole Logic** → *Disjointment*
- 5. **Rotation Logic** → *Triangle Paradigm*
- 6. **Position** → *Located Entity*
- 7. **Conclusion** → *Challenge*

### What is the module’s roadmap for formalizing motion?

Seven steps: classify motion in general (**Taxonomy**); define the two motion primitives — **Linear Motion** (location change) and **Rotational Motion** (orientation change); provide the logical machinery for reasoning about spatial containment while an entity moves (**Part-Whole Logic** — Disjointment); deep-dive on the trickiest rotational concept, relative left/right (**Rotation Logic** — Triangle Paradigm); formalize how an entity’s position is ultimately anchored (**Position** — Located Entity); and close with an honest assessment of how genuinely hard motion remains for machines (**Conclusion** — Challenge). This is the direct kinematic counterpart to Module D’s Time/Space agenda, built up separately and then re-merged via Motion on Slide 16.

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### Slides 5–7: Taxonomy of Motion

- **Distance:** Which distance has to be overcome?
- **Velocity:** How fast is the movement?
- **Precision:** How precise has the motion path and target location to be defined?
- **Complexity:** How irregular are path and rotation of the moving part?
- **Interaction:** How much interaction is necessary? (Predictability)
- **Intelligence:** How many intelligence do you need for the motion?
- **Mass** (*Slide 7, additional dimension*): How high is the mass of the moving part (also an energy aspect)? (*list left open with “...”*)
- **Visual (Slide 6):** A radar chart plotting **industrial, mobile, service, and humanoid robots** against the six core dimensions (1–5 scale).

### What are the Six (plus one) Dimensions of Motion, and why do they matter?

Distance and Velocity are the classic physical parameters (how far, how fast); Precision captures how tightly the path/endpoint must be controlled; Complexity captures how irregular the required path/rotation is; Interaction captures how much the motion must be renegotiated against an unpredictable environment; and Intelligence captures how much cognitive processing (planning, adaptation) the motion actually demands — tying the physical act of moving directly back to Module I’s Mind architecture. **Mass** is added as a candidate seventh dimension, explicitly flagged as both a mechanical (inertia/load) and energy-cost factor, with the list deliberately left open (“...”) to signal the taxonomy isn’t claimed to be exhaustive.

### How do different robot types compare across these dimensions?

Per the radar chart (Slide 6): an **industrial robot** scores high on Precision and Velocity but low on Interaction/Intelligence, since it blindly repeats a fixed, predictable path; a **humanoid robot** scores high across Complexity, Intelligence, and Interaction, since it must navigate unpredictable, human-populated environments with irregular, adaptive motion — often at the cost of raw Velocity/Precision. This is a practical engineering rubric: rather than treating “robotics”

as one monolithic problem, the six-dimension profile lets an engineer specify exactly which axes a given application actually demands.

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## Slides 8–10: Linear and Rotational Motion

### What is Linear Motion (Translation)?

**All points of an object move on parallel paths, in the same direction, and with the same speed** — a pure change of **location**, with nothing about the object’s internal orientation changing. The module anchors this to a **Location Reference Entity (LRE)**, directly reusing Module D’s Relative Space Logic apparatus (a Reference Entity plus reference points computing a Location Vector, see *Session D*, Slides 21–22).

### What is Rotational Motion?

**All points of an object rotate in a circle around one central axis** — a pure change of **orientation**. Points farther from the axis travel a greater linear distance per rotation than points closer to it, even though every point shares the same angular displacement. This corresponds to the **Orientation Vector** component of Module D’s same Relative Space Logic framework.

### Why does the distinction between Linear and Rotational motion matter?

Together they establish that any rigid body’s motion decomposes into exactly two independent components — matching precisely the 3+3 (Location + Orientation) six-degrees-of-freedom pose model from Module D (see *Session D*, Slide 20). In robotics/kinematics terms, this is the foundation for how any physical action, no matter how complex, is calculated: as a matrix-math mix of translation and rotation (e.g., via quaternions or Euler angles) applied to servo motors, not “magic energy.”

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## Slide 11: Part-Whole Logic — Transitivity, Disjointness/Exclusivity

- Worked hierarchy: Germany → Bavaria → Würzburg → Franz-Horn-Strasse → Cairo Building → {Staircase, Corridor, K.0.11} → K.0.11 → {North/East/South/West Wall} → Door is part of North Wall.
- **Transitivity (same time)**: if <A> is part of <B> and <B> is part of <C> then <A> is part of <C>. Example: **K.0.11 is part of Europe**.
- **Disjointness (same time)**: if <A> is part of <B> and !<C> is part of <B> then !<A> is part of <C>. Example: if X: Robert is part of K.0.11 and Y: Robert is part of Corridor then time(X) != time(Y).

### How does an AI reason about motion between spatial containers, rather than only tracking continuous coordinates?

Via two logical rules reused directly from earlier modules. **Transitivity** (Module G’s part-whole hierarchy, see *Session G*, Slide 21) lets the system chain arbitrarily long containment relations together — K.0.11 is part of the Cairo Building is part of Franz-Horn-Strasse is part of Würzburg is part of Bavaria is part of Germany is part of Europe, therefore K.0.11 is part of Europe. **Disjointness/Exclusivity** (directly reusing Module I’s Exclusivity axiom, see *Session I*, Slide 23) enforces that if K.0.11 and the Corridor are disjoint (non-overlapping) parts of the

same building, Robert cannot be part of both at the same time — if the record shows him in both, those observations *must* have occurred at different times (`time(X) != time(Y)`), meaning he **moved** between them. This is the precise logical mechanism for detecting motion purely from containment-change facts, without needing a continuous coordinate trace — exactly the reasoning a smart-building or indoor-tracking system would run, and exactly why a robot whose internal database claims it’s simultaneously in the Kitchen and the Garage must throw an error and trigger sensor recalibration, since that violates physical Disjointness.

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## Slide 12: Workshop “Rotation” (2 brainstorming prompts)

**What/where is left and what/where is right? Give a definition understandable by each human being without using any tools.**

This is the module’s deepest puzzle, deliberately left difficult here and resolved formally on the next two slides: left/right cannot be defined as fixed, absolute properties of the world at all — they only exist relative to an observer’s own orientation.

**What happens if you turn yourself 90 degrees and 180 degrees? Remains left and right the same?**

No — turning 180 degrees completely swaps what was “left” into “right” and vice versa (and a 90-degree turn maps what was “left/right” onto “front/back”). This proves left/right are not intrinsic, but strictly relative to the observer’s current rotational orientation — directly echoing Module I’s “WHERE is the View Point?” exercise (see *Session I*, Slide 22): relative spatial predicates are meaningless without an explicit, anchored observer, and for robotics/HRI, a robot cannot rely on a naively hard-coded “left = negative X” mapping, since that mapping breaks the instant the robot (or the speaker) rotates.

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## Slide 13: Rotation Logic — The Triangle Rotation Paradigm

- Three sequential triangle diagrams (A/B/C vertices), each rotated  $\sim 120^\circ$  from the last:
  - A is viewpoint  $\rightarrow$  direction (B is left to C).
  - C is viewpoint  $\rightarrow$  direction (A is left to B).
  - B is viewpoint  $\rightarrow$  direction (C is left to A).

**How is the Workshop’s left/right puzzle formally resolved?**

By proving that “left” and “right” are never an intrinsic property of two points/entities alone (e.g., B and C) — they are always a **three-way relation** requiring a third element: a viewpoint/direction of attention. As the same fixed triangle is viewed from three different vantage points (A, then C, then B in turn), the *same two physical vertices* swap which one is labeled “left” relative to the other — purely because the observing vantage point changed, with nothing about the triangle itself moving. This generalizes Module D’s Relative Space Logic model (a Reference Entity plus a front-facing vector, see *Session D*, Slide 22) into its most stripped-down, purely logical form: three points plus a rotating vantage.

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## Slide 14: Rotation Logic — What is Right? What is Left?

**What is Right?** → A Motion Concept: move our attention (vision, audio) to where the sun rises; when you then look in the direction where the sun is at its highest, you turn right (in the northern hemisphere).

**What Helps?** → Sun turns right (northern hemisphere); Clock hands turn right.

**What is Left?** → A Motion Concept: move our attention (vision, audio) to where the sun sets; when you then look in the direction where the sun is at its highest, you turn left (in the northern hemisphere).

**Challenge?** → Anchor points.

Having proven Left/Right require a viewpoint (Slide 13), this slide supplies a concrete, tool-free, universally learnable **anchor** for establishing that viewpoint: the sun’s daily apparent motion. Facing the sun’s daily zenith (its highest point, roughly south in the northern hemisphere), turning toward where it *rises* (east) is “right,” and turning toward where it *sets* (west) is “left” — and this exact convention is why clock hands (designed to mimic a sundial’s shadow) turn “clockwise”/“right.” The honestly-labeled **Challenge**, “Anchor points,” acknowledges that this entire convention depends on having a fixed reference (facing the solar zenith) to rotate from — and notably flips in the southern hemisphere, where the sun’s apparent motion reverses. This shows a seemingly arbitrary convention (why clocks go “clockwise”) is actually grounded in a genuine, physically observable, tool-free natural anchor — the same kind of natural-measurement-unit reasoning Module D used for solar day/year (see *Session D*) and Module H used for grounding names, now applied to the left/right convention itself.

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## Slide 15: Position — Located Entity

- A triangle (vertices A/B/C) viewed from a fixed viewpoint **X**: “**From my viewpoint X, A is right to B and C is right to B.**”

**How can both “A is right to B” and “C is right to B” be true at the same time without contradiction?**

Because, from one single, fixed viewpoint (X), these are two *separate* pairwise relations (X→A relative to B, and X→C relative to B) — not a claim that A and C occupy the same position relative to each other. A **Position** is only fully specified once expressed as a relation anchored to one consistent, explicitly stated viewpoint; multiple such relations can coexist without contradiction as long as they all share that same anchor. This is a precise illustration of why a “Located Entity” needs its position expressed relative to an *explicit* reference/viewpoint, exactly as formalized in Module D’s 6-parameter Relative Space Logic model (see *Session D*, Slide 22) — without stating the shared anchor (X) explicitly, the two claims would read as contradictory, when they are in fact perfectly consistent once the anchor is made explicit.

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## Slide 16: The Synthesis of Motion (Revisited)

- Time and space are mental concepts, that we use to describe what we perceive as motion.
- Motion welds time and space together.

- A complete description of motion only needs the temporal and the spatial dimensions.
- Diagram: **Entity** → **Motion** → **Time** [s] (*when? how long?*) & **Space** [m] (*where from? where to? how — path, rotation?*) → **Speed** [m/s].

#### How does this module’s climax relate to Module D’s own conclusion?

It is a near-verbatim restatement of Module D’s own climactic synthesis (see *Session D*, Slide 25), now serving as this module’s capstone claim, and directly cashing out the promise made by the module’s own subtitle (“Causer of Time and Space,” Slide 1): rather than Time and Space being pre-existing containers that Motion merely happens within, Motion is framed as the primary, perceived phenomenon, with Time and Space as the descriptive concepts derived from — and existing purely in service of — describing it. Formally, **Speed** = **Space** / **Time** ([m/s]), and a complete description of any motion needs nothing beyond these two dimensions. Together, Modules D and K describe the exact same Entity–Motion–Time–Space–Speed relationship from opposite starting points — Module D built Time and Space up as independent guarantors and then showed Motion welds them together; Module K starts from Motion itself and shows Time and Space are simply its descriptive byproducts — proving the tri-directional interdependency holds regardless of which concept is treated as logically prior.

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#### Slides 17–18: Conclusion

- Motion is the change of location and/or orientation of things in the physical world.
- We are all experts in motion when it comes to our body movement, which is based on the rotational motion of our joints.
- Motion in all its facets is very challenging for machines (Physical AI).
- One of the biggest challenges are obstacles that suddenly appear and “destroy” the action/motion plan which is just being executed (see road accidents).
- The left-right paradigm is an easy method to get closer to the way humans solve the motion challenge, but probably it is not the only one we use.

#### What are the module’s final takeaways?

Motion reduces exactly to the two primitives established in Sections 2–3: change of **location** and/or **orientation**. Humans are unconscious, effortless experts at motion — every human body movement ultimately reduces to the rotational motion of joints (arms, legs, fingers are all rotational actuators) — yet motion remains one of the hardest unsolved problems for machines (Physical AI), challenging across nearly every one of the Six Dimensions (Slide 5). The single biggest practical challenge singled out is **sudden, unpredicted obstacles** that invalidate an action plan already being executed (e.g., road accidents) — directly reflecting the Interaction/Predictability dimension from Slide 5, and echoing Module I’s point (see *Session I*, Slide 27) about needing continuous immediate/intermediate sensory feedback to detect and correct a failing plan mid-execution. Finally, the module explicitly hedges on completeness: the left-right paradigm (Slides 12–15) is offered as *one* useful, tractable method humans use to solve relative-orientation problems, not necessarily the *only* mechanism the human mind actually relies on — an honest acknowledgment that motion, despite feeling trivially easy to a body shaped by millions of years of evolved rotational-joint control, remains a genuinely unsolved frontier for Physical AI.

## Session L — Brain

(Note: *L.md* summarizes its 49 slides in condensed, grouped ranges rather than reproducing every individual Q&A matrix verbatim, as *H.md*–*J.md* did. Where the source groups many slides together without restating exact workshop wording, the questions below are framed to match the conceptual content actually given, rather than invented verbatim prompts.)

### Slide 1: Title and Administration

#### What is the point of the final module, “L Brain — Thinking Machine”?

To bring the entire eleven-module seminar full circle: having separately, rigorously defined every philosophical component of intelligence across the series — Reality (C), Time and Space (D), Information (E), Action (F), Concept (G), Memory (H), Mind (I), Language (J), and (per the agenda) Motion (K) — this final module bolts all of those components together into a single, comprehensive, literal system architecture: the **Digital Brain**, i.e., a genuine blueprint for a **Thinking Machine**.

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### Slides 2–4: Workshop and Agenda

- Header: **L Brain**
- Agenda Items:
  1. **Humans & Machines** → *Crowning Glory of Creation and Their Creations*
  2. **Thinking Theory** → *Source and Sink of Information and Action*
  3. **Brain Architecture** → *The Thought Machine*
  4. **Brain Screens** → *including Mind’s Eye*
  5. **Thought Window** → *Defined Information Exchange*
  6. **AI Spots** → *Machine Learning*
  7. **Conclusion** → *Strong Artificial Intelligence*

#### What is the final module’s roadmap for actually building an AGI?

Seven steps: (1) re-establish the mechanical relationship between **Humans & Machines** (Module A); (2) summarize the unified **Thinking Theory** that ties Information and Action together (Modules E/F); (3) provide the literal block diagram for the **Brain Architecture** — CPU, Memory, I/O; (4) define **Brain Screens** — the internal interfaces (including the Mind’s Eye) through which the brain’s modules communicate; (5) define the **Thought Window** — the defined channel of information exchange between those screens; (6) place **Machine Learning** (“AI Spots”) correctly within that architecture as a set of narrow utility components, not the seat of intelligence itself; and (7) conclude with the case for **Strong Artificial Intelligence** as a buildable, not merely theoretical, system.

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### Slides 5–10: Humans & Machines

- **Slide 6:** Humans are “Working Individuals.” Our basic instinct is to “Simplify Work.” We are intelligent because we have excellent sensors, brains, and musculoskeletal systems.
- **Slide 7:** Work connects People and Machines via the Division of Labor.
- **Slides 8–9:** A Machine is a “Work Performer.” A Work System brings humans and machines together via a Task, Tools, and a Process.

- **Slide 10:** Defines Intelligence (via the Duden dictionary): *The ability to think abstractly and reasonably and derive purposeful acting out of that.*

### Why does the final module re-open with Humans, Machines, and Intelligence — content already covered in Module A?

To re-anchor the seminar’s ultimate goal before presenting the final architecture: **to perform Work**. Per the seminar’s foundational premises (Module A, see *Session A*), humans build machines because our basic instinct is to *simplify our own labor*; a Machine, correctly understood, is simply a “Work Performer” (mechanical or non-mechanical); and Intelligence, per the Duden definition used throughout (Modules B and F), is the ability to think abstractly/reasonably and derive purposeful acting from that. The point of restating all this here, at the very end, is to make explicit that an AGI is not a “magic spirit” — it is just a **non-mechanic Work Performer**: a software system designed to autonomously complete tasks within a Work System (Actor/Goal/Tool/Process/Object), exactly as defined at the start of the entire seminar.

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### Slides 11–30: Thinking Theory (The Unified Model)

- **The Theory Triad (Slide 14):** Thinking Theory is the mathematical intersection of *Information Theory* and *Action Theory*.
- **The Missing Link (Slide 17):** “No operationalisable thinking theory = no thinking machine until now.”
- **The Elementary Thinkables (Slides 19–21):** The absolute base primitives of the system are **Entity**, **Time**, **Number**, **Space**, and **Name**.
- **Logic (Slides 22–24):** Entities are connected using strict Temporal and Spatial Logic.

### What is Thinking Theory, formally?

The mathematical **intersection of Information Theory (Module E) and Action Theory (Module F)** — i.e., a unified theory that doesn’t treat “what an agent knows” and “what an agent does” as separate concerns, but as two faces of a single underlying model. This directly echoes Module F’s Slide 23 triangle (Action  $\leftrightarrow$  Thought  $\leftrightarrow$  Information forming one continuous feedback loop, see *Session F*), now formalized as the seminar’s capstone theoretical claim.

### Why hasn’t a genuine “thinking machine” (AGI) been built yet?

Per the module’s central diagnosis: “**No operationalisable thinking theory = no thinking machine until now.**” The bottleneck was never raw compute, data volume, or engineering effort (echoing Module C’s Slide 4 observation about 50+ years of failed attempts to model the world) — it was the *absence of a precise, implementable definition of what a thought actually is*. Computer science has been trying to build brains without first rigorously specifying the object primitives and logical operations a “thought” is made of — which is exactly the gap the rest of this module (and, retrospectively, the entire seminar) claims to close.

### What are the “Elementary Thinkables” — the absolute base primitives needed to build a thinking machine?

Five: **Entity** (Module C/G’s delimitable atomic unit), **Time** (Module D’s mental concept for ordering movement), **Number** (Module G’s primitive concept for quantity), **Space** (Module D’s mental concept for delimitability), and **Name** (Module G/H’s phonetic/textual pointer). The claim is that engineers do not need to code “consciousness” or anything mystical — they need only implement these five object primitives and the logical operators that bind them together.

## How are these Elementary Thinkables (Entities) connected to each other?

Via strict **Temporal Logic** (implemented concretely as **StartTime/EndTime** properties, per Module H’s **TABLE ASSOCIATION** schema) and **Spatial Logic** (implemented concretely as **X, Y, Z** coordinate properties, per Module D’s **Space Logic**) — the same two logic families identified as “Basic/Complex Logics” throughout Module I (see *Session I*), now confirmed as the literal binding mechanism for every Entity in the final architecture.

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## Slides 31–36: Brain Architecture (The Block Diagram)

- **Slide 32:** A 3D cube — the **L Brain** — containing four modular blocks:
  1. **I Mind** (Central Logic Unit / CPU)
  2. **H Memory** (Persistence / Hard Drive)
  3. **J Language Center** (I/O Protocol)
  4. **K Motion Center** (Kinematics / Actuators)
- **Slide 34:** The Memory block is partitioned into specialized databases (Episode, Values, Motion, Language, Knowledge, Planning).
- **Slide 35:** The Mind block is partitioned into specific execution threads (Understanding, Reasoning, Learning, Logic, Initiating).

## What is the literal, final system architecture of a Digital Brain?

Four cleanly decoupled modules inside one overall container (the “L Brain” cube): **I Mind** acting as the Central Logic Unit/CPU (Module I, see *Session I*); **H Memory** acting as the persistence layer/hard drive (Module H, see *Session H*); a **J Language Center** acting as the I/O protocol handler (Module J, see *Session J*); and a **K Motion Center** handling kinematics/actuators (Module K). Crucially, this proves an AGI is **not** one massive, monolithic, black-box neural network — it is a highly modular, decoupled system where the logic engine (Mind) is completely separated from the database (Memory), and external communication (Language) and physical movement (Motion) are independent API-style drivers around that core.

## How is the Memory block internally organized?

Into specialized, partitioned databases: **Episode, Values, Motion, Language, Knowledge, and Planning** — a direct, expanded restatement of Module H’s Memory Segments taxonomy (Short-term/Long-term, Episode/Reality/Individual Reality/Value/Action Plans/Motion Concepts/Speech Concepts, see *Session H*), now mapped explicitly onto the final architecture’s storage layer.

## How is the Mind block internally organized?

Into specific execution threads: **Understanding, Reasoning, Learning, Logic, and Initiating** — an exact restatement of Module I’s five functional sections (see *Session I*), now presented as literal parallel/sequential processing threads inside the CPU block of the final architecture.

## Why does this modularity matter?

Because it directly resolves two well-known failure modes of current large language models: **catastrophic forgetting** and **hallucination**. Today’s LLMs attempt to compress Memory, Logic, and Language into one homogeneous matrix of learned weights, with no clean separation between “what is known,” “how it is validated,” and “how it is expressed.” This architecture instead mandates that these be cleanly separated into distinct microservices — a persistent,

structured database (Memory); a rule-governed reasoning engine (Mind); and independent translation layers (Language, Motion) — so that, for instance, updating stored knowledge doesn’t require retraining the entire reasoning engine, and expressing a thought in language doesn’t risk corrupting the underlying logic that produced it.

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### Slides 37–43: Brain Screens and Thought Windows

- **Slide 38:** Sensors (Vision, Hearing) push data into the Brain. Actuators (Moving, Speaking) pull data out.
- **Slide 39:** Worked example — vision detects a spider.
- **Slide 40:** The internal visual protocol is called **GREBTL** (Generic Role-Entity-Based Thought Language).
- **Slide 42:** “Screens” are the internal interfaces (APIs) between the brain sections.
- **Slide 43:** Thoughts are *Association Networks* displayed on a screen — not a sequential channel. The “Mind’s Eye” is simply the part of the screen the brain is currently focused on.

### How do the Brain’s separate modules (Mind, Memory, Language, Motion) actually communicate with one another?

Through “**Screens**” — internal interfaces functioning as APIs between brain sections. Sensors (Vision, Hearing) push new data onto a screen; actuators (Moving, Speaking) pull commands off a screen to execute them. Critically, per Slide 43, what gets displayed on a screen is not a sequential stream of text or tokens — it is a full **Association Network** (Module G’s graph structure), and the “**Mind’s Eye**” (already introduced in Module I, see *Session I*) is simply whichever part of that screen the brain’s active attention is currently focused on — the same “highlighted node” concept from Module I’s castle-graph example, now generalized into the architecture’s core communication primitive.

### What protocol governs the data passed between these internal screens?

**GREBTL** (Generic Role-Entity-Based Thought Language) — the same formal notation introduced in Module G to describe derived-concept algebra (see *Session G*, Slide 23), now confirmed as the actual internal payload format used across the whole Brain architecture, rather than a one-off illustrative tool.

### Walk through a concrete example: how does seeing a spider lead to running away, in this architecture?

Per Slide 39’s worked example: the **Vision sensor** detects a spider and renders it as an association network onto the visual **Screen**. The **I Mind** reads that screen, cross-checks it against the **Value Memory** partition (per Slide 34 — stored values like “spiders are bad,” Module H’s segmented Value Memory), and, upon detecting a mismatch between the perceived state and the agent’s values (Module F’s action-trigger condition), immediately generates and renders an **Action Plan** (“Run Away!”) onto the Motion **Screen**. The **K Motion Center** then reads that screen and fires the corresponding physical leg-motor commands. This is a complete, concrete walkthrough of the entire seminar’s action loop (Perception → Value comparison → Action Plan → Execution) implemented as a literal data flow between four decoupled modules.

### Why does it matter that the payload between modules is a graph (GREBTL) rather than raw natural-language text?

Because it defines the **data contract** between AGI microservices precisely: the internal payload is a standardized, structured Knowledge Graph, not a sequential string — meaning downstream modules (Motion, Language) receive unambiguous, already-parsed relational data rather than having to re-interpret free text, avoiding the exact kind of semantic ambiguity Module J’s Semantics section (see *Session J*) showed natural language is prone to (e.g., syntactically valid but semantically hollow tautologies).

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## Slides 44–45: AI Spots (Machine Learning)

- Header: **6. AI Spots**
- Subtitle: **Using Machine Learning**
- Core Visual (Slide 45): the Brain block diagram, with traditional “AI” (Machine Learning models) acting as isolated utility scripts:
  - **Visualiser/Editor AI**: Translates screen language to diagrams.
  - **Real Time Detector AI**: Translates raw sensor data into screen language.
  - **Translator AI**: Translates Natural Language to internal Screen Language.
  - **Value Improver AI**: Updates the Value database.

## Is Machine Learning (Deep Learning / LLMs) the same thing as Strong AI?

**No** — this is the module’s deliberate, explicit paradigm shift. Deep Learning models are cast as “**Spots**”: dumb, narrowly-specialized utility scripts operating only at the *edges* of the true AGI, not as the seat of intelligence itself. Concretely: a **Real Time Detector AI** (a conventional neural net) translates raw sensor data (pixels, audio waveforms) into the internal Screen Language (GREBTL); a **Translator AI** converts external Natural Language (Module J) into that same internal Screen Language and back; a **Visualiser/Editor AI** renders Screen Language into human-readable diagrams; and a **Value Improver AI** updates the Value Memory database over time. In every case, ML is doing translation/perception work at the system’s boundary — the actual “Intelligence,” per this architecture, is the **deterministic graph-logic engine** (the Mind block) sitting at the center, not any of the neural-network components.

## Why does this reframing matter?

It resolves the seminar’s recurring tension (first raised in Module A/B, see *Session A* and *Session B*) between narrow/weak AI and Strong AI/AGI: rather than treating large neural networks as an incomplete-but-scaling path toward AGI, this final architecture treats them as permanently narrow, specialized components — genuinely useful, but categorically distinct from the symbolic, graph-based reasoning core that this module claims actually constitutes “thinking.”

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## Slides 46–49: Conclusion

- The brain is a machine that processes real information in the form of representation networks.
- The architecture must not necessarily be complex.
- The key lies in the different logics that have to be used and bound together to drive that machine.
- **Slide 48**: A workshop asking why the billion-euro “Human Brain Project” failed.

### What are the seminar’s final takeaways?

The brain — biological or artificial — is fundamentally a machine that processes real information in the form of **representation networks** (Modules C, E, G’s Association Networks). The resulting architecture, despite integrating eleven modules’ worth of theory, **does not have to be immensely complex**: the key is not raw scale or biological fidelity, but correctly identifying and binding together the **right set of logics** (Naming, Placeholder, Abstraction, Transitivity, Exclusivity, Temporal, Spatial — per Module I, see *Session I*) that drive the machine.

### Why did the billion-euro “Human Brain Project” fail?

The module poses this as a final workshop question, and its own framework strongly implies the answer: that project attempted to achieve artificial general intelligence by simulating **biological neurons** at massive scale and fidelity, rather than by building a **logic-based, symbolic architecture** of the kind specified across this entire seminar (Entities, Associations, Concepts, bound by Temporal/Spatial/Exclusivity logic). The implied lesson is that replicating the brain’s *wetware* substrate is neither necessary nor sufficient — what matters is replicating its underlying **software architecture** (a structured graph-logic engine), which can, per this seminar’s central thesis, be implemented today using standard computer science tools: graph databases, logic gates, and APIs — not exotic neuromorphic hardware or biologically-faithful neuron simulation.

### What is the seminar’s ultimate message to the software engineers in the room?

Optimism, grounded in engineering feasibility rather than speculation: you do **not** need to build a perfect biological replica of a human brain to achieve Strong AI. The biological brain is simply “wetware” running a specific, discoverable software architecture (Entity/Concept/Association Network + Time/Space/Exclusivity Logic + a decoupled Mind/Memory/Language/Motion block structure). If that same software architecture is faithfully replicated using standard, already-available computer science tools, Strong Artificial Intelligence is presented as a buildable target today — not a distant, uncertain future milestone.